

# A FINITELY PRESENTED GROUP THAT IS NOT MF

## ANONYMOUS

**ABSTRACT.** A countable discrete group is MF if it admits unitary matrix models that are asymptotically multiplicative in operator norm and separate every nontrivial element from the identity; equivalently, if it embeds in the unitary group of a quotient  $\prod_n M_{d_n}(\mathbb{C})/\bigoplus_n M_{d_n}(\mathbb{C})$ . We construct an explicit finitely presented group  $E$  that is not MF, refuting the conjecture that every countable group is MF. The reduced group  $C^*$ -algebra of  $E$  is a separable, stably finite  $C^*$ -algebra that is not MF.

The mechanism is a rigidity theorem for property-(T) subgroups: if  $t\Gamma t^{-1} \subseteq \Gamma$ , then in operator-norm matrix models conjugation by  $t$  preserves the Hilbert–Schmidt asymptotic commutant of  $\Gamma$ . In  $E$  this forces a commutator  $u$  to converge to 1 in normalized Hilbert–Schmidt norm. On the other hand, an explicit Clifford-group quotient makes the central involution  $w = u^2$  equal to  $-1$ . Hence every matrix model of  $E$  maps  $w$  to 1. The construction runs from any finitely presented property-(T) group with a proper injective self-embedding;  $E$  is the instance built on  $\mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$ .

The homomorphism proving  $w \neq 1$  maps  $E$  onto a second group  $W$ , a Clifford extension of a linear group. We prove that  $W$  is finitely generated, sofic, and not MF. Since sofic groups are hyperlinear, neither soficity nor hyperlinearity implies the MF property. The reduced group  $C^*$ -algebra of  $W$  is in addition exact.

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## 1 INTRODUCTION

Can every countable discrete group be realized by matrices, if multiplication is only required to hold up to operator-norm errors that vanish along a sequence of finite dimensions? Shulman calls the positive assertion the *MF conjecture* [Sh], and no group was known to fail it [Sc, Sh]. We prove that the answer is no: there is an explicit finitely presented group, on eight generators and forty-one relators, that is not MF. Residually finite groups satisfy the conjecture through their finite quotients, as do LEF groups [CDE] and, by quasidiagonality [TWW], all amenable groups. Recent work establishes it for many amalgamated free products [Sc, Sh, GKEMP]. The corresponding  $C^*$ -algebra question is older. Blackadar and Kirchberg, who introduced MF algebras, remarked that no stably finite separable  $C^*$ -algebra was known not to be MF [BK, CDE]. Their natural candidate was  $C_{\text{red}}^*(F_2)$ , which Haagerup and Thorbjørnsen later proved to be MF [HT]. The negative solution of the Connes embedding problem subsequently supplied abstract stably finite non-MF algebras [JNVWY, FGH]. No countable group was known whose reduced or maximal group  $C^*$ -algebra fails to be MF [Sh].

These questions sit at the endpoint of a program of finitary approximation. In his 2018 ICM address, Thom asked for which Schatten  $p$ -norms there exist non-approximable groups [Th, Question 3.11]. The case  $p = 2$  was settled by De Chiffre–Glebsky–Lubotzky–Thom [DGLT], the cases  $1 < p < \infty$  by Lubotzky–Oppenheim [LuO], and the case  $p = 1$  by Bachner–Dogon–Lubotzky [BDL]. That left the Hamming metric (soficity), the Hilbert–Schmidt norm (hyperlinearity), and the operator norm (MF). The Hamming case was disproved in 2026: the first nonsofic group was constructed by OpenAI [OAI], Fournier-Facio then gave a torsion-free example [FFF], and Kun–Thom developed further wreath-product examples [KT26]. One-sided conjugation of a property-(T) subgroup into itself, the mechanism driving those arguments, is also the starting point of the present one. Soficity concerns permutations in the Hamming metric, and MF concerns unitaries in operator norm, so the operator-norm setting requires a different rigidity argument. We obtain it from a one-sided transport theorem for the Kazhdan fixed spaces of the adjoint actions; Appendix B gives a quantitative finite-stage version. This paper settles the operator-norm case by one explicit finitely presented group; it also proves that soficity does not imply MF and produces an explicit reduced group  $C^*$ -algebra that is stably finite but not MF. Of the three problems, only hyperlinearity remains open.

In the language of approximate representations, a countable discrete group  $G$  is *MF* if, for every finite subset  $F \subseteq G$  and every  $\varepsilon > 0$ , there is a unitary

matrix model  $V : G \rightarrow \mathcal{U}(r)$  that is multiplicative within  $\varepsilon$  in operator norm on  $F$  and separates distinct elements of  $F$  by operator-norm distance at least 1. There is an equivalent corona formulation, the one in which the notion was introduced by Carrión–Dadarlat–Eckhardt and is used by Shulman and Bachner–Dogon–Lubotzky [CDE, Sh, BDL]. Let  $(d_n)_{n \in \mathbb{N}}$  be a strictly increasing sequence of positive integers and let

$$(1) \quad \mathcal{Q} = \prod_n M_{d_n}(\mathbb{C}) / \bigoplus_n M_{d_n}(\mathbb{C})$$

be the associated matrix quotient: bounded sequences of matrices modulo sequences converging to 0 in operator norm. We call any such corona algebra a *norm matrix corona*. Following Blackadar–Kirchberg [BK], a separable  $C^*$ -algebra is *MF* if it embeds into some algebra of the form (1); the group  $G$  is MF if there is an injective homomorphism

$$G \longrightarrow \mathcal{U}(\mathcal{Q}).$$

The two formulations are equivalent (Proposition 1.1). We shall use the equivalent coordinate model

$$(2) \quad \mathcal{U}_{\text{cor}}((d_n)) := \prod_n \mathcal{U}(M_{d_n}(\mathbb{C})) / \{(u_n) : \|u_n - 1\| \rightarrow 0\}.$$

The denominator is normal, and polar correction gives a canonical isomorphism

$$(3) \quad \kappa_{(d_n)} : \mathcal{U}_{\text{cor}}((d_n)) \xrightarrow{\cong} \mathcal{U}(\mathcal{Q}),$$

proved in Lemma 2.2. Thus one may equivalently embed  $G$  into  $\mathcal{U}_{\text{cor}}((d_n))$ .

**Proposition 1.1** (equivalent MF formulations). *For every countable group, the  $C^*$ -corona definition above is equivalent to the unitary-sequence definition. The latter is equivalent to operator-norm local models with separation constant 1, and allowing arbitrary positive dimensions is equivalent to requiring the dimensions to be strictly increasing.* [Lean: GroupApproximation.ManuscriptExactWrappers.manuscriptMFDefinitionEquivalences]

The proof is deferred, together with the precise formulation of the operator-norm local models, to Appendix A. Its three ingredients are an explicit tensor-power amplification in one direction, the weak-MF-to-corona implication in the other, and a cumulative-block dimension normalization.

Hereafter “MF” means this property: the operator-norm notion of Carrión–Dadarlat–Eckhardt, in the form used by Shulman, with no additional requirement that matrix models recover the regular representation, its norms, or its trace. Stronger conventions exist. In [GKEMP] the approximating maps must also recover the canonical trace and the left-regular norm on every group-ring element. The purely matricial field (PMF) convention of [MdlS, Definition 1.2] requires finite-dimensional representations whose group-ring norms converge to the regular norms.

Throughout,  $[g, h] = ghg^{-1}h^{-1}$  for group elements, while for matrices  $[x, y] = xy - yx$ ; the arguments always make the reading unambiguous.

**The result and the mechanism** Theorem A produces an explicit group  $E$  on eight generators  $v_1, v_2, v_3, x, y, z, t, c$  with forty-one relators, containing a designated central involution  $w$ , such that  $w \neq 1$  and yet every homomorphism from  $E$  to the unitary group of a norm matrix corona sends  $w$  to 1. In particular no such homomorphism is injective, and  $E$  is not MF.

The underlying mechanism concerns a general configuration: a property-(T) group  $\Lambda$ , a homomorphism  $\iota: \Lambda \rightarrow H$ , and elements  $t, c \in H$  with

$$t \iota(\Lambda) t^{-1} \subseteq \iota(\Lambda), \quad [c, \iota(\Lambda)] = 1.$$

An operator-norm almost representation of  $H$  acts by conjugation on the Hilbert–Schmidt space of each matrix block, and operator-norm control keeps this adjoint action approximately multiplicative. Property (T) supplies the Kazhdan projection  $P$  onto the subspace of vectors almost fixed by the adjoint action of  $\iota(\Lambda)$ ; the one-sided inclusion gives  $P \leq V P V^*$  for the adjoint unitary  $V$  of  $t$ ; and the ambient  $C^*$ -ultraproduct is finite — it contains no proper isometries — so  $P = V P V^*$ : conjugation by  $t$  preserves the Hilbert–Schmidt asymptotic commutant of  $\iota(\Lambda)$  (Theorem 3.1). At each fixed matrix dimension the same step is quantitative, with an almost-fixed spectral subspace in place of the Kazhdan projection and the equal-rank reversal estimate in place of finiteness (Appendix B).

Consequently  $d = t c t^{-1}$  asymptotically commutes with  $\iota(\Lambda)$ , so every element of the *compression defect* — the normal subgroup

$$N_{\text{comp}} = \langle\langle [d, \iota(\lambda)] : \lambda \in \Lambda \rangle\rangle_H$$

— satisfies  $\|U_n(g) - 1\|_2 \rightarrow 0$  in every operator-norm almost representation  $(U_n)$  of  $H$ . The obstruction theorems convert this tracial vanishing into operator-norm conclusions: every finite normal subgroup of  $H$  contained in  $N_{\text{comp}}$ , and more generally every normal property-(T) subgroup contained in it, lies in the kernel of every corona representation (Theorems 4.2 and 4.6).

For  $E$ , the base subgroup is the image of a group  $\mathcal{B}$  with property (T), proved intrinsically from the displayed presentation, and the marked word is the square of the defect commutator  $u = [d, v_1]$ , so  $\{1, w\}$  is a finite central subgroup of  $N_{\text{comp}}$ . What remains is nontriviality of  $w$ , and only there does the Clifford algebra enter: an explicit Clifford-group quotient of  $E$  makes  $d$  and  $v_1 d v_1^{-1}$  anticommuting involutions, so  $w = u^2$  maps to  $-1$ . In one line:

$$\text{Kazhdan transport: } u \sim_2 1; \quad \text{Clifford model: } w = u^2 = -1.$$

Section 11.3 proves a second application of the same one-sided compression, under a commuting-orbit hypothesis: when an involution is centralized by the compressed copy and its conjugates under the property-(T) subgroup commute, every corona representation identifies the whole conjugation orbit, and the MF radical is computed by the quotient by the resulting collapse defect (Theorems 11.14 and 11.18).

Two groups play distinct roles. The finitely presented group  $E$  is the main example: it is not MF, and we do not determine whether  $E$  itself is sofic or

hyperlinear (Section 12). The homomorphism proving  $w \neq 1$  maps  $E$  onto a second, finitely generated group  $W$ , a Clifford extension of a linear group, and Theorem E proves that  $W$  is sofic — hence hyperlinear — and not MF. Soficity is established for  $W$ , not for  $E$ . On the algebra side, the reduced group  $C^*$ -algebra of  $E$  is the stably finite non-MF example (Theorem D); the reduced algebra of  $W$  is in addition exact.

**Operator norm versus tracial approximation** The argument is specific to operator norm: the transport theorem requires the adjoint actions to be controlled in operator norm on the Hilbert–Schmidt spaces. Hilbert–Schmidt-small defects alone do not provide that control, so the proof does not give a hyperlinear obstruction.

**Inputs and formal verification** The proof of Theorem A is self-contained. Property (T) of the base is proved from an exact rational spectral certificate, an explicit surjection verified relator by relator, and a fixed-point bridge (Proposition 7.2). The Clifford witness and the finite-dimensional obstruction are proved directly.

Each gray Lean tag names a theorem in the accompanying Lean 4 development with the same statement.

**Relation to prior work** The closest conceptual analogue is the recent work on nonsofic groups. One-sided conjugation of a property-(T) subgroup and propagation from its centralizer appear in [OAI, KT26], building on rigidity results of Kun and Kun–Thom [Ku16, KT19]. Related centralizer rigidity for sofic embeddings is proved in [AT]. Those results concern Hamming or tracial approximations. Here the adjoint actions stay on the Hilbert–Schmidt spaces of the matrix blocks, and finiteness prevents a one-sided compression of the Kazhdan fixed space from being proper. In quantitative form this is the equal-rank reversal estimate for explicit finite-stage spectral subspaces (Appendix B). The complementary finite-group corner then converts that rigidity into a fixed nontrivial element sent to the identity by every corona representation. The Clifford central extension supplies the element, whose nontriviality is proved independently.

Slofstra’s constructions supply close precedents for the Clifford group, distinguished central involution, shift, and HNN doubling map [Slo18, Section 2 and Remark 3.3]. Negative-eigenspace compression appears in Slofstra–Vidick [SV, proof of Proposition 2.7], and Bachner–Dogon–Lubotzky develop operator-norm rounding, compression, and polar-correction procedures in detail [BDL, Lemmas 2.2–2.3, Proposition 2.4, and the proof of Proposition 1.5]. That local procedure enters the finite-normal corner reduction below. The transport step itself is Theorem 3.1.

The analytic passage from operator norm to the Hilbert–Schmidt conjugation representation belongs to the  $\pi \otimes \bar{\pi}$  framework of Bekka and Bekka–Valette [Be, BV]. Dadarlat uses normalized finite-rank projections as almost

invariant Hilbert–Schmidt vectors in an operator-norm approximation argument [Dad, Lemma 3.18 and Proposition 3.19]. These are the direct analytic precedents for the conjugation step used here.

Slofstra’s quotients by elements invisible to exact or approximate finite-dimensional representations precede the MF-radical language used below [Slo17, Definitions 2.5–2.7]. Fritz’s marked  $\text{BS}(2, 3)$  example provides additional context for the cyclic comparison [Fri, SV].

Central extensions and rigidity of approximate representations also drive the Frobenius obstruction of De Chiffre–Glebsky–Lubotzky–Thom [DGLT]. In the proof of Proposition 1.6, Bachner–Dogon–Lubotzky treat the finite-central case by a character-isotypic corner, and reduce the general finite-normal case to it by passing to the finite-index centralizer [BDL]. The finite-normal criterion here combines the corresponding coordinate-corner construction with the one-sided Kazhdan conjugation and does not assume that the quotient is hyperlinear. Related finite-target approximation and extension results for weak soficity appear in [GR, GI].

Carrión–Dadarlat–Eckhardt provide the group-MF framework used here. Through Abels’ group, they also provide an amenable MF group that is not residually finite [CDE, Section 2.14 and Corollary 2.18]. The cyclic comparison addresses the narrower question of what happens to the marked conjugation configuration when property (T) is removed.

Finally, failure of the MF property for a group  $C^*$ -algebra does not by itself prevent the group from embedding into a corona: the group property requires only a faithful canonical image in some MF quotient of  $C_{\max}^*(E)$ . For  $E$ , the nontrivial element  $w$  lies in the kernel of every corona representation.

## 2 NOTATION AND CORONA LEMMAS

All groups are discrete and countable. Property (T) is the textbook complex-unitary property. A  $C^*$ -homomorphism is unital only when this is stated. In particular, an MF embedding may be nonunital, as is customary for embedding a unital algebra as a  $C^*$ -subalgebra. If  $e: A \rightarrow B$  is an injective, possibly nonunital  $*$ -homomorphism between unital  $C^*$ -algebras, then  $u \mapsto e(u) + (1 - e(1))$  is an injective homomorphism  $\mathcal{U}(A) \rightarrow \mathcal{U}(B)$ ; this complement correction is used below. For a unital  $C^*$ -algebra  $A$  we write  $\mathcal{U}(A)$  for its unitary group. On  $M_r(\mathbb{C})$  we use the normalized trace  $\text{tr}_r$ , the operator norm  $\|\cdot\|$ , and the normalized Hilbert–Schmidt norm  $\|x\|_2 = \text{tr}_r(x^*x)^{1/2}$ . For coordinate sequences we write  $a \sim_2 b$  when  $\|a_n - b_n\|_2 \rightarrow 0$ . We use without comment:  $\|x\|_2 \leq \|x\|$ ;  $|\text{tr}_r(x)| \leq \|x\|$ ; and  $\|uxv\|_2 = \|x\|_2$  for unitaries  $u, v$ . For a nonempty finite set  $Y$ , write  $M_Y(\mathbb{C}) = \mathbb{C}^{Y \times Y}$  for the corresponding indexed matrix algebra.

Given a sequence  $(d_n)_{n \in \mathbb{N}}$  of positive integers, we write

$$\mathcal{Q}((d_n)_{n \in \mathbb{N}}) = \prod_{n \in \mathbb{N}}^{\ell^\infty} M_{d_n}(\mathbb{C}) / \bigoplus_{n \in \mathbb{N}}^{c_0} M_{d_n}(\mathbb{C}),$$

the quotient of the bounded  $C^*$ -product by the ideal of sequences with  $\|x_n\| \rightarrow 0$ . We call any such algebra a *norm matrix corona*. We also write

$$\mathcal{U}_{\text{cor}}((d_n)_{n \in \mathbb{N}}) = \prod_{n \in \mathbb{N}} \mathcal{U}(M_{d_n}(\mathbb{C})) / \{(u_n) : \|u_n - 1\| \rightarrow 0\}.$$

Write  $q: \prod_{n \in \mathbb{N}}^{\ell^\infty} M_{d_n}(\mathbb{C}) \rightarrow \mathcal{Q}$  for the quotient map of a norm matrix corona.

The following lifting facts are standard perturbation theory; see Loring [Lo] for a systematic treatment.

**Lemma 2.1** (lifting). *Every unitary  $u \in \mathcal{Q}$  lifts to a sequence of unitaries.*

[Lean: GroupApproximation.ManuscriptExactWrappers.manuscriptUnitaryLifting]

*Proof.* Lift  $u$  to an arbitrary bounded sequence  $(x_n)$ . Since  $q((x_n))$  is unitary, the one-sided Gram defect satisfies  $\|x_n^* x_n - 1\| \rightarrow 0$ . Once  $\|x_n^* x_n - 1\| \leq \frac{1}{2}$ , the square Gram matrix  $x_n^* x_n$  is invertible, hence so is  $x_n$ , and the polar correction  $u_n = x_n (x_n^* x_n)^{-1/2}$  is unitary. Continuous functional calculus gives the quantitative estimate

$$\|u_n - x_n\| \leq 2 \|x_n\| \|x_n^* x_n - 1\|.$$

The sequence  $(x_n)$  is bounded, so the right-hand side tends to 0. Setting  $u_n = 1$  at the finitely many remaining indices therefore gives a unitary sequence with  $\|u_n - x_n\| \rightarrow 0$ .  $\square$

**Lemma 2.2** (unitaries in the quotient). *For every sequence  $(d_n)$ , the map*

$$\kappa_{(d_n)}: \mathcal{U}_{\text{cor}}((d_n)) \longrightarrow \mathcal{U}(\mathcal{Q}), \quad [(u_n)] \longmapsto q((u_n)),$$

*is a canonical group isomorphism.* [Lean: GroupApproximation.ManuscriptExactWrappers.manuscriptUnitaryCoronaEquivalence]

*Proof.* The map is well defined, and its kernel consists exactly of the unitary sequences satisfying  $\|u_n - 1\| \rightarrow 0$ , so it is injective. It is surjective by Lemma 2.1: every unitary in  $\mathcal{Q}$  has a unitary lift  $(u_n)$ , whose class in (2) maps to it.  $\square$

A *corona representation* of a group  $H$  is a homomorphism  $H \rightarrow \mathcal{U}(\mathcal{Q})$ , and it *detects*  $g \in H$  — separates  $g$  from the identity — when it does not send  $g$  to 1. An *operator-norm asymptotic representation* of  $H$  is a sequence of maps  $U_n: H \rightarrow \mathcal{U}(d_n)$  such that  $\|U_n(g)U_n(h) - U_n(gh)\| \rightarrow 0$  for all  $g, h \in H$ . Lemma 2.2 canonically identifies a corona representation with a homomorphism into  $\mathcal{U}_{\text{cor}}((d_n))$ , which we call its coordinate model. A countable group is *MF* if it admits an injective corona representation for a strictly increasing dimension sequence; by the dimension-amplification equivalence, allowing an arbitrary positive dimension sequence gives the same class of groups. Taking  $g = h = 1$  gives  $\|U_n(1) - 1\| \rightarrow 0$  automatically, since  $U_n(1)$  is unitary. Coordinate unitary lifts of a corona representation of a countable group form exactly such a sequence.

The maximal group  $C^*$ -algebra  $C_{\text{max}}^*(H)$  is used only for the final assertions of Theorem A. Appendix C constructs it and proves its universal property.

## 3 ONE-SIDED KAZHDAN TRANSPORT

The proof rests on one principle, which appears in three finite settings:

$$\begin{array}{ccc} X \subseteq Y, |X| = |Y| & V \subseteq W, \dim V = \dim W & \begin{array}{c} p \leq q, p \sim q \\ \text{in a finite } C^*\text{-algebra} \end{array} \\ \Downarrow & \Downarrow & \Downarrow \\ X = Y & V = W & p = q. \end{array}$$

In the right-hand column,  $p \sim q$  is Murray–von Neumann equivalence and *finite* is meant in the  $C^*$ -sense: no proper isometries. The middle implication is used in Theorem B: the commutant of the base is carried into itself by conjugation, and equal finite dimension forces equality. For matrix models, the right-hand implication gives the proof printed below: property (T) supplies a Kazhdan projection  $P$  in a norm ultraproduct, the one-sided conjugation gives  $P \leq VPV^*$ , and finiteness forces equality. The same argument runs quantitatively at each single coordinate, with an almost-fixed spectral subspace in place of the Kazhdan projection and the equal-rank reversal estimate in place of finiteness. That finite-stage proof is given in Appendix B.

Recall from Section 2 the notions of corona representation, detection, and operator-norm asymptotic representation, and the notation  $a_n \sim_2 b_n$  for  $\|a_n - b_n\|_2 \rightarrow 0$  in the normalized Hilbert–Schmidt norm.

**Theorem 3.1** (Kazhdan transport of Hilbert–Schmidt asymptotic commutants). *Let  $\Gamma$  and  $H$  be groups, let  $\Gamma$  have property (T), let  $\iota: \Gamma \rightarrow H$  be a homomorphism, let  $s \in H$ , and let  $(d_n)$  be a sequence of positive integers. Suppose*

$$s\iota(\Gamma)s^{-1} \subseteq \iota(\Gamma).$$

*Let  $U_n: H \rightarrow \mathcal{U}(d_n)$  be an operator-norm asymptotic representation. If  $(x_n)$  is uniformly operator-norm bounded and asymptotically centralizes  $\iota(\Gamma)$  in normalized Hilbert–Schmidt norm,*

$$\|x_n U_n(\iota(\gamma)) - U_n(\iota(\gamma)) x_n\|_2 \rightarrow 0 \quad (\gamma \in \Gamma),$$

*then the conjugated sequence does as well:*

$$\|[U_n(s)x_n U_n(s)^*, U_n(\iota(\gamma))]\|_2 \rightarrow 0 \quad (\gamma \in \Gamma).$$

[Lean: GroupApproximation.KazhdanAsymptoticCommutant.manuscriptKazhdanTransport]

*Proof.* Suppose the conclusion fails: there are  $\gamma_0 \in \Gamma$ ,  $\delta > 0$ , and an infinite set  $I \subseteq \mathbb{N}$  with  $\|[U_n(s)x_n U_n(s)^*, U_n(\iota(\gamma_0))]\|_2 \geq \delta$  for  $n \in I$ . Fix a free ultrafilter  $\omega$  on  $\mathbb{N}$  with  $I \in \omega$ .

*The adjoint model.* Regard  $M_{d_n}(\mathbb{C})$  as the Hilbert space  $K_n = L^2(M_{d_n}(\mathbb{C}), \text{tr}_{d_n})$  and let each group element act on it by conjugation,  $\text{Ad } U_n(g) \xi = U_n(g) \xi U_n(g)^*$ . Operator-norm almost multiplicativity of  $U_n$  makes  $g \mapsto \text{Ad } U_n(g)$  an operator-norm asymptotic representation on these Hilbert spaces; this is the finite-stage form of the  $\pi \otimes \bar{\pi}$  construction.

*The ultraproduct.* Let  $K_\omega$  be the Hilbert-space ultraproduct of the  $K_n$  along  $\omega$  and let  $B_\omega = \prod_\omega B(K_n)$  be the norm ultraproduct, acting on  $K_\omega$ ;



write  $[\cdot]_\omega$  for classes in either. Asymptotic multiplicativity makes

$$\pi: H \longrightarrow \mathcal{U}(B_\omega), \quad \pi(g) = [\text{Ad } U_n(g)]_\omega,$$

a homomorphism. The algebra  $B_\omega$  is *finite*. Indeed, suppose  $\sigma = [\sigma_n]_\omega$  satisfies  $\sigma^* \sigma = 1$ . Then  $\|\sigma_n^* \sigma_n - 1\| \rightarrow 0$  along  $\omega$ , and each  $\sigma_n$  acts on a finite-dimensional space. Polar correction therefore replaces  $\sigma_n$  by a unitary  $w_n$  with  $\|\sigma_n - w_n\| \rightarrow 0$ . Hence  $\sigma = [w_n]_\omega$  is unitary.

*The Kazhdan projection.* Property (T) provides a finite symmetric generating set  $S \subseteq \Gamma$  with  $1 \in S$  and a Kazhdan constant  $\kappa \in (0, 1]$  for the pair  $(S, \kappa)$ . Let  $h$  be the Hermitian part of  $\frac{1}{|S|} \sum_{s' \in S} \pi(\iota(s'))$  and let  $\text{Fix} \subseteq K_\omega$  be the subspace of vectors fixed by every  $\pi(\iota(\gamma))$ . Both  $\text{Fix}$  and its orthocomplement are invariant under each unitary  $\pi(\iota(s'))$ , hence under  $h$ . On  $\text{Fix}$ ,  $h = 1$ . The restriction of  $\pi \circ \iota$  to  $\text{Fix}^\perp$  has no nonzero invariant vector, so a unit vector  $\xi \in \text{Fix}^\perp$  satisfies  $\|\pi(\iota(s'))\xi - \xi\| \geq \kappa$  for some generator. For that generator  $\text{Re}\langle \pi(\iota(s'))\xi, \xi \rangle \leq 1 - \kappa^2/2$ , while every other term is at most 1. Averaging gives  $\langle h\xi, \xi \rangle \leq 1 - \kappa^2/(2|S|)$ . Consequently

$$\text{sp}(h) \subseteq \left[-1, 1 - \frac{\kappa^2}{2|S|}\right] \cup \{1\},$$

and the spectral projection  $P$  of  $h$  at the isolated point 1 belongs to  $B_\omega$  by continuous functional calculus and projects onto  $\text{Fix}$ .

*One-sided compression and finiteness.* Let  $V = \pi(s)$  and  $Q = VPV^*$ . Since  $s\iota(\Gamma)s^{-1} \subseteq \iota(\Gamma)$ , every vector of  $\text{Fix}$  is fixed by each  $\pi(s\iota(\gamma)s^{-1})$ . On the other hand,  $\eta$  is fixed by all  $\pi(s\iota(\gamma)s^{-1})$  exactly when  $V^*\eta \in \text{Fix}$ . Hence

$$\text{ran } P = \text{Fix} \subseteq V \text{Fix} = \text{ran } Q, \quad \text{that is,} \quad P \leq Q.$$

The element  $r = V^*Q$  satisfies  $r^*r = Q$  and  $rr^* = V^*QV = P$ . Since  $r^*r = Q$ , we have  $r = rQ$ . Since  $\text{ran } r = \text{ran } P \subseteq \text{ran } Q$ , we also have  $r = Qr$ . Hence the cross terms in  $\sigma^* \sigma$  vanish for  $\sigma = r + (1 - Q)$ , and  $\sigma^* \sigma = Q + (1 - Q) = 1$ :  $\sigma$  is an isometry. Finiteness of  $B_\omega$  makes  $\sigma$  unitary, and  $\sigma \sigma^* = P + (1 - Q)$  forces

$$Q = P.$$

*Conclusion.* Let  $\xi = [\xi_n]_\omega \in K_\omega$ , where  $\xi_n \in K_n$  is the vector of  $x_n$ ; the uniform operator-norm bound makes  $\xi$  well defined. By unitary invariance of the normalized Hilbert–Schmidt norm, the commutator hypothesis says exactly that each  $\pi(\iota(\gamma))$  fixes  $\xi$ . Thus  $\xi \in \text{Fix}$ . Since  $Q = P$ , we obtain

$$V\xi \in V \text{Fix} = \text{ran } Q = \text{ran } P = \text{Fix}.$$

The vector of  $U_n(s)x_n U_n(s)^*$  is  $\text{Ad } U_n(s) \xi_n$ , so, again by unitary invariance,

$$\|[U_n(s)x_n U_n(s)^*, U_n(\iota(\gamma))]\|_2 = \|(\text{Ad } U_n(\iota(\gamma)) - 1) \text{Ad } U_n(s) \xi_n\| \longrightarrow 0$$

along  $\omega$ , for every  $\gamma \in \Gamma$ . Taking  $\gamma = \gamma_0$  contradicts  $I \in \omega$ . Since the failing subsequence was arbitrary, the full sequence converges.  $\square$

*Remark 3.2* (the finite-stage proof). The ultrafilter hides the rate. Appendix B proves the theorem a second time, quantitatively at each single coordinate  $n$ . There, an explicit almost-fixed spectral subspace of the averaged adjoint action replaces the Kazhdan projection, and the equal-rank reversal estimate replaces finiteness of  $B_\omega$ . That proof yields an explicit rate in terms of  $\kappa$ ,  $|S|$ , the uniform bound on  $(x_n)$ , and the multiplicative defects of  $U_n$ .

The transport theorem holds at every Hilbertian scale, not only at the dimension normalization.

**Theorem 3.3** (Kazhdan transport at every scale). *Let  $\Gamma$ ,  $H$ ,  $\iota$ ,  $s$ , and  $(U_n)$  be as in Theorem 3.1, and let  $(w_n)$  be nonnegative weights. Suppose there is  $C$  with  $\text{Tr}(x_n^* x_n) \leq C w_n$  for all  $n$ , and that for every  $\gamma \in \Gamma$  and every  $\varepsilon > 0$ , eventually*

$$\text{Tr} |x_n - U_n(\iota(\gamma)) x_n U_n(\iota(\gamma))^*|^2 \leq \varepsilon w_n.$$

*Then  $U_n(s) x_n U_n(s)^*$  and  $U_n(s)^* x_n U_n(s)$  satisfy both conditions as well.*

[Lean: GroupApproximation.ScaledKazhdanTransport.scaled\_transport\_both]

*Proof.* Give each coordinate space  $M_{d_n}(\mathbb{C})$  with  $w_n > 0$  the inner product  $\langle x, y \rangle = \text{Tr}(y^* x)/w_n$ ; conjugation by a unitary is unitary for every weight, both hypotheses and both conclusions concern only these renormalized norms, and an index with  $w_n = 0$  forces  $x_n = 0$ . The ultraproduct proof of Theorem 3.1 applies verbatim: the mass bound defines the ultraproduct vector, the adjoint actions remain operator-norm almost multiplicative on the weighted Hilbert spaces, and finiteness reverses the one-sided compression of the fixed-space projection in both directions. The formal development follows the finite-stage route of Appendix B at the weight  $(w_n)$ .  $\square$

The dimension weight  $w_n = d_n$  recovers Theorem 3.1, with the uniform operator-norm bound relaxed to mass boundedness. The collapse proof of Theorem 11.14 uses a rank weight, which normalizes a defect at its own scale.

The transported object can also be an intertwiner between two different asymptotic representations; the adjoint sector of Theorem 3.1 is the diagonal case.

**Theorem 3.4** (transport of asymptotic intertwiners). *Let  $\Gamma$ ,  $H$ ,  $\iota$ , and  $s$  be as in Theorem 3.1, and let  $U_n^1: H \rightarrow \mathcal{U}(d_n^1)$  and  $U_n^2: H \rightarrow \mathcal{U}(d_n^2)$  be operator-norm asymptotic representations. Let  $(x_n)$  be uniformly operator-norm bounded  $d_n^1 \times d_n^2$  matrices with*

$$\frac{\text{Tr} |x_n - U_n^1(\iota(\gamma)) x_n U_n^2(\iota(\gamma))^*|^2}{d_n^1 + d_n^2} \longrightarrow 0 \quad (\gamma \in \Gamma).$$

*Then  $U_n^1(s) x_n U_n^2(s)^*$  satisfies the same convergence for every  $\gamma \in \Gamma$ .*

[Lean: GroupApproximation.IntertwinerKazhdanTransport.manuscriptIntertwinerTransport]

*Proof.* The block sum  $W_n = U_n^1 \oplus U_n^2$  is an operator-norm asymptotic representation, because block-diagonal operator norms are maxima. The off-diagonal embedding of  $x_n$  into the block space has operator norm  $\|x_n\|$ ;

its commutator with  $W_n(\iota(\gamma))$  is the off-diagonal embedding of the displayed intertwining defect, whose normalized Hilbert–Schmidt norm on the block space carries the denominator  $d_n^1 + d_n^2$ ; and conjugation by  $W_n(s)$  acts on the embedded corner by  $x_n \mapsto U_n^1(s) x_n U_n^2(s)^*$ . Theorem 3.1, applied to  $W_n$ , transports the embedded sequence. The same reduction applied to Theorem 3.3 gives the intertwiner transport at every weight. [Lean: GroupApproximation.IntertwinerKazhdanTransport.scaled\_intertwiner\_transport]  $\square$

Fixed tensors of type  $(p, q)$  are Hilbert–Schmidt intertwiners from the  $q$ th to the  $p$ th tensor power. Tensor powers of unitaries are exactly multiplicative and satisfy  $\|A^{\otimes p} - B^{\otimes p}\| \leq p \|A - B\|$ , so tensor powers of an operator-norm asymptotic representation are again operator-norm asymptotic representations, and Theorem 3.4 applies to every pair of them: one-sided compression of a Kazhdan image creates no asymptotic invariant tensors of any fixed type, at any weight. The tensor-power instantiation is not part of the formal development.

#### 4 THE COMPRESSION DEFECT AND ITS OBSTRUCTION THEOREMS

The transport theorem has a group-theoretic output. We fix the configuration it needs once and for all.

**Definition 4.1** (Kazhdan conjugation datum). Let  $H$  be a countable group. A *Kazhdan conjugation datum* in  $H$  is a tuple  $(\Lambda, \iota, t, c)$ , where  $\Lambda$  is a countable group with property (T),  $\iota: \Lambda \rightarrow H$  is a group homomorphism, not assumed injective, and  $t, c \in H$ , subject to

- (M1)  $t \iota(\lambda) t^{-1} \in \iota(\Lambda)$  for every  $\lambda \in \Lambda$ ;
- (M2)  $c \iota(\lambda) = \iota(\lambda) c$  for every  $\lambda \in \Lambda$ .

Write  $d := tct^{-1}$ . The *defect subgroup* of the datum is the normal closure

$$N_{\text{comp}} = \langle\langle [d, \iota(\lambda)] : \lambda \in \Lambda \rangle\rangle_H \trianglelefteq H.$$

[Lean: GroupApproximation.ManuscriptExactWrappers.manuscriptMarkedKazhdanPattern]

Both conditions are relations among generators, so a datum can be read off a presentation. The relations force  $c$  to commute with  $\iota(\Lambda)$  and force  $d$  to commute with the compressed copy  $t\iota(\Lambda)t^{-1}$ , but they do not constrain the commutators  $[d, \iota(\lambda)]$ ; the defect subgroup  $N_{\text{comp}}$  — the *compression defect* of the datum — is their normal closure. The theorems of this section show that every finite normal subgroup of  $H$  contained in  $N_{\text{comp}}$ , and more generally every normal property-(T) subgroup contained in it, lies in the kernel of every corona representation of  $H$ .

**Theorem 4.2** (finite-normal obstruction criterion). *Let  $H$  be a countable group with a Kazhdan conjugation datum  $(\Lambda, \iota, t, c)$ , and let  $F \trianglelefteq H$  be a finite normal subgroup with  $F \subseteq N_{\text{comp}}$ . Then every corona representation  $\Theta: H \rightarrow \mathcal{U}(\mathcal{Q})$  satisfies  $\Theta(f) = 1$  for all  $f \in F$ .* [Lean: GroupApproximation.ManuscriptExactWrappers.manuscriptFiniteNormalObstructionCriterion]

*Proof. Tracial vanishing of the defect.* Let  $(U_{g,n})$  be coordinate unitary lifts of any operator-norm almost representation of  $H$  (in particular, of a corona representation), and put

$$K_2 = \{g \in H : \|U_{g,n} - 1\|_2 \rightarrow 0\}.$$

Unitary invariance and asymptotic multiplicativity make  $K_2 \trianglelefteq H$ . Since  $c$  centralizes  $\iota(\Lambda)$ , its lifts belong to the Hilbert–Schmidt asymptotic commutant of  $\iota(\Lambda)$ . Applying Theorem 3.1 to  $x_n = U_{c,n}$  puts the lifts of  $d = tct^{-1}$  there as well. Therefore  $[d, \iota(\lambda)] \in K_2$  for every  $\lambda \in \Lambda$ ; since  $K_2$  is normal, it contains the normal closure of these commutators:  $N_{\text{comp}} \subseteq K_2$ , and so  $F \subseteq K_2$ . We call the elements of  $K_2$  *Hilbert–Schmidt invisible* in the given almost representation; every element of the defect subgroup is thus invisible in every operator-norm almost representation of  $H$ .

*The coordinate Reynolds corner.* Suppose a corona representation  $\Theta$  is nontrivial on  $F$ ; choose coordinate unitary lifts  $(V_{g,n})$  of  $\Theta$  and form the coordinate Reynolds averages

$$p_n = \frac{1}{|F|} \sum_{f \in F} V_{f,n}.$$

Because the lifts are asymptotically multiplicative and  $F$  is finite,  $(p_n)$  is asymptotically self-adjoint and idempotent, and normality of  $F$  makes it asymptotically central for all of  $\Theta(H)$ . Spectral rounding at  $1/2$  of the Hermitian part of  $p_n$  produces genuine coordinate projections; let  $q_n$  be their complements. Nontriviality of  $\Theta$  on  $F$  forces  $q_n \neq 0$  along infinitely many coordinates — otherwise the averages converge to 1 and  $\Theta|_F$  is trivial. Retain exactly those coordinates and relabel them by  $\mathbb{N}$ . Compress every lift to the corner  $q_n M_{d_n}(\mathbb{C}) q_n \cong M_{r_n}(\mathbb{C})$  with  $r_n = \text{rank } q_n > 0$ , and polar-correct the compressed matrices. The result is an operator-norm almost representation  $(W_{g,n})$  of  $H$  on the nonzero corner blocks. Since the corner is the complement of the rounded average, the finite average vanishes there:

$$(*) \quad \left\| \sum_{f \in F} W_{f,n} \right\| \rightarrow 0 \quad \text{in operator norm.}$$

*The contradiction.* The tracial vanishing applies to the corner almost representation  $(W_{g,n})$  itself: every  $f \in F$  satisfies  $\|W_{f,n} - 1\|_2 \rightarrow 0$ . Since  $F$  is finite, a diagonal selection of coordinates makes this uniform over  $F$ . Passing to stages at which all  $|F|$  distances are below  $1/(n+1)$  gives

$$\sum_{f \in F} W_{f,n} \rightarrow |F|1 \quad \text{in normalized Hilbert–Schmidt norm,}$$

each block carrying its own normalized trace. This contradicts  $(*)$ , because operator-norm convergence dominates normalized Hilbert–Schmidt convergence. Hence no corona representation is nontrivial on  $F$ .  $\square$

The Reynolds-corner argument used only one property of  $N_{\text{comp}}$ : every element is Hilbert–Schmidt invisible in every operator-norm almost representation of  $H$ . Any normal subgroup with that tracial-vanishing property therefore obstructs its finite normal subgroups in exactly the same way; Theorem 4.5 exploits this.

For a two-element central subgroup  $F = \{1, \varepsilon\}$  the Reynolds average is  $p_n = \frac{1}{2}(1 + V_{\varepsilon,n})$ , and the corner is the negative spectral corner of the sign; compression to it gives a model in which  $\varepsilon = -1$  exactly. This case is the form in which Theorem 4.2 is applied to the explicit group of Theorem A, and it deserves its own statement.

**Theorem 4.3** (central-sign criterion). *Let  $H$  and  $\Gamma$  be countable groups, let  $\Gamma$  have property (T), let  $\iota: \Gamma \rightarrow H$  be a homomorphism, and suppose*

$$t\iota(\Gamma)t^{-1} \subseteq \iota(\Gamma), \quad [c, \iota(\Gamma)] = 1.$$

*Put  $d = tct^{-1}$  and  $u = [d, \iota(a)]$  for some  $a \in \Gamma$ . If*

$$\varepsilon = u^2 \neq 1, \quad \varepsilon^2 = 1, \quad \varepsilon \in Z(H),$$

*then every homomorphism from  $H$  to a norm matrix corona maps  $\varepsilon$  to the identity, and  $H$  is not MF.* [Lean: GroupApproximation.KazhdanCompressionCore.manuscriptCentralSignCriterion]

*Proof.* The tuple  $(\Gamma, \iota, t, c)$  is a Kazhdan conjugation datum in  $H$ , and  $u = [d, \iota(a)]$  is one of the displayed generators of  $N_{\text{comp}}$ , so  $\varepsilon = u^2 \in N_{\text{comp}}$ . The subgroup  $F = \{1, \varepsilon\}$  is finite, and normal because  $\varepsilon$  is central. Theorem 4.2 therefore gives  $\Theta(\varepsilon) = 1$  for every corona representation  $\Theta$  of  $H$ . Since  $\varepsilon \neq 1$ , no corona representation is injective, and  $H$  is not MF.  $\square$

**4.1 The intrinsic defect** The transport estimate applies to every one-sided conjugator of a subgroup, not only to the element  $t$ . For a subgroup  $L \leq H$ , let  $G_{\text{comp}}(L)$  be generated by the elements  $s$  satisfying  $sLs^{-1} \subseteq L$  — the *one-sided conjugators* of  $L$  — and put

$$(4) \quad \mathfrak{D}(H, L) = \left\langle \left\langle [gzg^{-1}, \ell] : g \in G_{\text{comp}}(L), z \in C_H(L), \ell \in L \right\rangle \right\rangle_H,$$

the *intrinsic defect* of  $L$  in  $H$ . It depends only on the pair  $(H, L)$ , not on a choice of datum.

**Corollary 4.4** (transport along all one-sided conjugators). *Let  $(r_n)$  and  $V_{g,n} \in \mathcal{U}(r_n)$  be an operator-norm asymptotic representation of  $H$ , let  $\Lambda$  have property (T), let  $\iota: \Lambda \rightarrow H$  be a homomorphism, and put  $L = \iota(\Lambda)$ . Suppose that  $(x_n \in M_{r_n}(\mathbb{C}))$  is uniformly bounded in operator norm and belongs to the Hilbert–Schmidt asymptotic commutant of  $L$ :*

$$\left\| x_n - V_{\iota(\lambda),n} x_n V_{\iota(\lambda),n}^* \right\|_2 \longrightarrow 0 \quad (\lambda \in \Lambda).$$

*Then every  $g \in G_{\text{comp}}(L)$  acts in both directions on that asymptotic commutant. Explicitly, both*

$$V_{g,n} x_n V_{g,n}^* \quad \text{and} \quad V_{g,n}^* x_n V_{g,n}$$

satisfy the preceding convergence for every  $\lambda \in \Lambda$ . [Lean: GroupApproximation.KazhdanAsymptoticCommutant.compressionGroup\_transport\_both]

*Proof.* For a one-sided conjugator  $s$ , Theorem 3.1 gives forward transport. Reverse transport follows from the same proof: the equality  $P = VPV^*$  obtained there gives  $V^*PV = P$  as well, so the fixed subspace is preserved by conjugation in both directions. In the finite-stage proof of Appendix B, the equal-rank reversal estimate reverses the one-sided containment of the spectral subspaces; the same argument then runs with the adjoint unitary, with the displacement and concentration steps unchanged. The elements acting in both directions form a subgroup of  $H$ , and hence contain the subgroup generated by all one-sided conjugators, namely  $G_{\text{comp}}(L)$ .  $\square$

**Theorem 4.5** (the intrinsic defect and the MF radical). *Let  $H$  and  $\Gamma$  be countable groups, let  $\Gamma$  have property (T), let  $\iota: \Gamma \rightarrow H$  be a homomorphism, and put  $L = \iota(\Gamma)$ . If  $F \trianglelefteq H$  is finite and  $F \leq \mathfrak{D}(H, L)$ , then every homomorphism from  $H$  to every norm matrix corona maps every element of  $F$  to the identity.* [Lean: GroupApproximation.KazhdanAsymptoticCommutant.manuscriptCompressionRadical]

*Proof.* Let  $(U_{h,n})$  be coordinate lifts of an operator-norm almost representation of  $H$  and put  $K_2 = \{h : \|U_{h,n} - 1\|_2 \rightarrow 0\}$ , a normal subgroup of  $H$ . If  $z \in C_H(L)$ , then  $(U_{z,n})$  lies in the asymptotic commutant of  $L$ , and Corollary 4.4 places  $(U_{gzg^{-1},n})$  there for every  $g \in G_{\text{comp}}(L)$ . Hence each generator  $[gzg^{-1}, \ell]$ ,  $\ell \in L$ , displayed in (4) belongs to  $K_2$ ; since  $K_2 \trianglelefteq H$ , it contains their normal closure:  $\mathfrak{D}(H, L) \leq K_2$ . Every element of  $\mathfrak{D}(H, L)$  is therefore Hilbert–Schmidt invisible in every operator-norm almost representation of  $H$ , and the coordinate Reynolds corner of Theorem 4.2 applies verbatim to any finite normal  $F \leq \mathfrak{D}(H, L)$ .  $\square$

**4.2 The normal-Kazhdan form of the obstruction** Finiteness of  $F$  enters Theorem 4.2 only through the Reynolds corner. A finite group has property (T), and its Reynolds average is its fixed-space average; for an arbitrary normal property-(T) subgroup, the spectral projection of a Kazhdan average takes its place. The resulting form of the obstruction has no finiteness, centrality, or torsion hypothesis; it is used in Section 12.

**Theorem 4.6** (normal-Kazhdan obstruction). *Let  $H$  be a countable group with a Kazhdan conjugation datum  $(\Lambda, \iota, t, c)$ , and let  $K \trianglelefteq H$  be a normal subgroup with property (T) such that  $K \subseteq N_{\text{comp}}$ . Then every corona representation  $\Theta: H \rightarrow \mathcal{U}(\mathcal{Q})$  satisfies  $\Theta(k) = 1$  for all  $k \in K$ .* [Lean: GroupApproximation.ManuscriptExactWrappers.manuscriptNormalKazhdanObstruction]

Taking  $K = F$  finite recovers Theorem 4.2, since finite groups have property (T): the whole group with tolerance 1 is a Kazhdan pair, by orbit averaging. [Lean: GroupApproximation.hasKazhdanPropertyT\_of\_finite] The two printed proofs differ in which corner they cut. In Theorem 4.6,  $K$  may be infinite, noncentral, and torsion-free.

*Proof.* The containment  $N_{\text{comp}} \subseteq K_2$  from the proof of Theorem 4.2 uses no finiteness, so every element of  $K$  is Hilbert–Schmidt invisible in every operator-norm almost representation of  $H$ . It remains to replace the Reynolds corner by a corner adapted to  $K$ ; the spectral projection of a Kazhdan average provides it.

Suppose a corona representation  $\Theta$  is nontrivial on  $K$ . Its image  $\bar{H} = \Theta(H)$  is a countable subgroup of  $\mathcal{U}(\mathcal{Q})$ , hence MF. Proposition 1.1 therefore provides, for every finite subset of  $\bar{H}$  and every tolerance, a matrix model multiplicative within the tolerance that separates distinct elements by *operator-norm* distance at least 1. Choosing models along an exhaustion of  $\bar{H}$  gives an operator-norm asymptotic representation  $(V_{g,n})$  of  $\bar{H}$  in which every fixed nontrivial element eventually stays at operator-norm distance at least 1 from the identity. Operator-norm separation is all such models provide: at growing dimension it places no lower bound on normalized Hilbert–Schmidt distances, and the corner cut below must manufacture the missing tracial separation. The datum and the subgroup pass to their images under  $\Theta$ : normality and property (T) are preserved, and  $\bar{K} = \Theta(K)$ , nontrivial by assumption, lies in the defect subgroup of the image datum. So it suffices to derive a contradiction for  $\bar{H}$ .

*The moving spectral subspace.* Fix a finite symmetric generating Kazhdan set  $S \subseteq \bar{K}$  with constant  $\varepsilon_0$ , at each coordinate let  $h_n$  be the Hermitian part of the average of the models of  $S$ , and cut the spectrum of  $h_n$  at  $\theta = 1 - \varepsilon_0^2/(4|S|)$ . Let  $q_n$  be the spectral projection of  $h_n$  for  $[-1, \theta]$ : the *moving spectral subspace*. On its complement — the top spectral subspace — every model of every element of  $\bar{K}$  is close to the identity in operator norm, by the Kazhdan-pair displacement estimate. If  $q_n = 0$  for all large  $n$ , every element of  $\bar{K}$  would satisfy  $\|V_{k,n} - 1\| \rightarrow 0$ , contradicting the operator-norm separation of any fixed nontrivial element of  $\bar{K}$ . So the moving subspace is nonzero along an infinite set of coordinates, which we retain.

*The tracial deficit on the corner.* Equip each retained corner block  $q_n M_{d_n}(\mathbb{C}) q_n$  with its renormalized trace. By the definition of the cut,  $q_n h_n q_n \leq \theta q_n$ , so the average over  $S$  of the real parts of the renormalized traces of the compressed models is at most  $\theta$ . Hence at every retained coordinate some generator in  $S$  has compressed model with real renormalized trace at most  $\theta$ , and therefore normalized Hilbert–Schmidt distance at least  $\sqrt{2(1-\theta)} - o(1)$  from the corner identity after polar correction; passing to a subsequence fixes a single such generator  $s_0$ . The spectral cutoff has converted operator-norm separation into a definite tracial deficit on its own corner.

*Compression.* Normality makes the cut asymptotically central: conjugation by an ambient model carries the  $S$ -average to the average of a conjugate generating set of  $\bar{K}$  with the same gap, whose top spectral subspace agrees asymptotically with the original — both consist of the almost- $\bar{K}$ -fixed vectors — so  $\|[q_n, V_{g,n}]\| \rightarrow 0$  for every  $g \in \bar{H}$ . Compressing every model to the moving corner and polar-correcting yields an operator-norm asymptotic



representation of all of  $\bar{H}$  on nonzero blocks, in which  $s_0$  keeps a definite normalized Hilbert–Schmidt distance from the identity.

*The contradiction.* The tracial vanishing applies verbatim to the corner representation:  $\bar{K}$  lies in the defect subgroup of the image datum, so every element of  $\bar{K}$  — in particular  $s_0$  — is Hilbert–Schmidt invisible on the corner, contradicting the tracial deficit of  $s_0$ . The uniform coordinate scheduling that runs these estimates simultaneously is the stage selection of Theorem 4.2.  $\square$

The proofs above use the conjugation datum through one property only: every element of the defect subgroup is Hilbert–Schmidt invisible in every operator-norm almost representation. The next theorem takes that invisibility as its hypothesis.

**Theorem 4.7** (abstract invisibility obstruction). *Let  $H$  be a countable group and let  $D \leq H$  be a subgroup such that for every operator-norm almost representation of  $H$ , the induced homomorphism into the tracial ultraproduct of its blocks, along any ultrafilter refining the cofinite filter, maps every element of  $D$  to the identity. Let  $K \triangleleft H$  be a normal subgroup with property (T) such that  $K \leq D$ . Then every corona representation  $\Theta: H \rightarrow \mathcal{U}(\mathcal{Q})$  satisfies  $\Theta(k) = 1$  for all  $k \in K$ .* [Lean: GroupApproximation.ManuscriptExactWrappers.manuscriptAbstractNormalKazhdanObstruction]

*Proof.* The proof of Theorem 4.6 used its datum only to make every element of the defect subgroup Hilbert–Schmidt invisible in every operator-norm almost representation; here that invisibility is the hypothesis. Suppose a corona representation  $\Theta$  is nontrivial on  $K$ . As there, the image  $\bar{H} = \Theta(H)$  is countable and MF, and Proposition 1.1 provides an operator-norm asymptotic representation of  $\bar{H}$  that separates every fixed nontrivial element. Every operator-norm almost representation of  $\bar{H}$  composes with  $\Theta$  to one of  $H$ , so every element of  $\Theta(D)$  inherits the invisibility hypothesis, while  $\bar{K} = \Theta(K) \leq \Theta(D)$  is normal, Kazhdan, and nontrivial. The moving spectral subspace, the tracial deficit, and the compression step of the proof of Theorem 4.6 apply verbatim to  $\bar{K}$ , and the contradiction step uses the inherited invisibility in place of the defect containment.  $\square$

Applied to the intrinsic defect, Theorem 4.7 gives the normal-Kazhdan companion of Theorem 4.5:

**Corollary 4.8** (normal-Kazhdan form of the intrinsic radical). *Let  $H$  and  $\Gamma$  be countable groups, let  $\Gamma$  have property (T), let  $\iota: \Gamma \rightarrow H$  be a homomorphism, and put  $L = \iota(\Gamma)$ . If  $K \trianglelefteq H$  is a normal subgroup with property (T) and  $K \leq \mathfrak{D}(H, L)$ , then every homomorphism from  $H$  to every norm matrix corona maps every element of  $K$  to the identity.* [Lean: GroupApproximation.ManuscriptExactWrappers.manuscriptIntrinsicNormalKazhdanRadical]

*Proof.* The proof of Theorem 4.5 shows that every element of  $\mathfrak{D}(H, L)$  is Hilbert–Schmidt invisible in every operator-norm almost representation of  $H$ . Theorem 4.7 with  $D = \mathfrak{D}(H, L)$  concludes.  $\square$



The subgroup generated by *all* normal property-(T) subgroups of  $\mathfrak{D}(H, L)$  therefore also lies in the kernel of every corona representation. [Lean: GroupApproximation.ManuscriptExactWrappers.manuscriptIntrinsicNormalKazhdanPart]

## 5 THE KAZHDAN–CLIFFORD CONSTRUCTION

Given  $\Gamma, \alpha, a$  as in the theorem below, the construction admits a presentation with one redundant generator  $\varepsilon$ : declare it to be a central involution and impose

$$(5) \quad \boxed{\varepsilon = u^2, \quad u = [tct^{-1}, a].}$$

Rigidity forces  $u \sim_2 1$ , whereas the Clifford model makes  $u^2 = -1$ . The theorem below uses the Tietze-eliminated version: since  $d = tct^{-1}$  is an involution,  $u^2 = [d, ada^{-1}]$ , and centrality of  $u^2$  conversely forces  $(u^2)^2 = 1$ .

**Theorem 5.1** (Kazhdan–Clifford construction). *Let  $\Gamma$  be a finitely presented group with property (T), let  $\alpha: \Gamma \hookrightarrow \Gamma$  be an injective endomorphism, and choose  $a \in \Gamma \setminus \alpha(\Gamma)$ . Adjoin elements  $t, c$  and impose*

$$t\gamma t^{-1} = \alpha(\gamma), \quad c^2 = 1, \quad [c, \gamma] = 1 \quad (\gamma \in \Gamma).$$

*Put*

$$d = tct^{-1}, \quad u = [d, a], \quad w = u^2 = [d, ada^{-1}],$$

*and impose centrality of  $w$ . More explicitly, for a finite generating set  $S_\Gamma$  it is enough to impose the conjugation and commutation relations for  $s \in S_\Gamma$ , together with*

$$[w, s] = 1 \quad (s \in S_\Gamma), \quad [w, t] = [w, c] = 1.$$

*Thus  $w$  commutes with a generating set of the entire resulting group. The group  $E(\Gamma, \alpha, a)$  is finitely presented, the canonical map  $\Gamma \rightarrow E(\Gamma, \alpha, a)$  is injective, and  $w$  is a nontrivial central involution. Every homomorphism*

$$E(\Gamma, \alpha, a) \longrightarrow \mathcal{U}(\mathcal{Q})$$

*to the unitary group of every norm matrix corona maps  $w$  to 1. Consequently  $E(\Gamma, \alpha, a)$  is not MF.* [Lean: GroupApproximation.KazhdanCliffordConstruction.kazhdanCliffordConstruction]

*Proof.* Finite presentability follows from a relative finite presentation: start with the free product of  $\Gamma$  and the free group on  $t, c$ , and quotient by the finitely many displayed relations for a finite generating set of  $\Gamma$ .

The relation  $w^2 = 1$  is redundant. Put  $b = ada^{-1}$ . Both  $d$  and  $b$  are involutions,  $w = [d, b] = (db)^2$ , and direct cancellation gives

$$(6) \quad dwd^{-1} = w^{-1}.$$

Centrality also gives  $dwd^{-1} = w$ , whence  $w = w^{-1}$  and  $w^2 = 1$ .

For nontriviality, realize the ascending HNN extension

$$G = \langle \Gamma, t \mid t\gamma t^{-1} = \alpha(\gamma) \rangle$$

as a mapping telescope. Let  $\Gamma_\infty$  be the direct limit of the system  $\Gamma \xrightarrow{\alpha} \Gamma \xrightarrow{\alpha} \cdots$ ; injectivity of  $\alpha$  passes to the limit, so the level-zero map embeds  $\Gamma$  in  $\Gamma_\infty$ . Applying  $\alpha$  levelwise induces an injective endomorphism  $\sigma$  of  $\Gamma_\infty$ . It is surjective, hence an automorphism, because an element at level  $k$  equals  $\sigma$  of the same element at level  $k+1$ . Put  $G = \Gamma_\infty \rtimes_\sigma \mathbb{Z}$ , with stable letter  $t$  implementing  $\sigma$ ; then  $t\gamma t^{-1} = \alpha(\gamma)$  for  $\gamma \in \Gamma$ . Write  $X = G/\Gamma$  for the coset space of the level-zero copy. Let  $G$  permute the Clifford generators  $(e_x)_{x \in X}$  and map  $c$  to  $e_\Gamma$ . Then

$$d \mapsto e_{t\Gamma}, \quad ada^{-1} \mapsto e_{at\Gamma}.$$

The two cosets are distinct: equality would give  $t^{-1}at \in \Gamma$ , equivalently  $a \in \alpha(\Gamma)$ . Hence the two Clifford generators anticommute and

$$w \mapsto (e_{t\Gamma}e_{at\Gamma})^2 = -1.$$

The Clifford sign is central, so the map respects every defining relation. It follows simultaneously that  $w \neq 1$  and that the canonical map of  $\Gamma$  is injective.

Finally,  $w = u^2$  is a nontrivial central involution. By the central-sign criterion (Theorem 4.3), every corona representation maps  $w$  to the identity. Since  $w \neq 1$ , none is faithful.  $\square$

The element  $a$  plays two roles. Analytically,  $a \in \Gamma$ , so fixed-space transport forces the displaced lamp to commute with its image in finite-dimensional norm models. Geometrically,  $a \notin \alpha(\Gamma)$ , so  $a$  moves that lamp to a different HNN coset, where Clifford anticommutation produces  $-1$ .

*Remark 5.2* (a small-cancellation realization). Fournier-Facio has pointed out (personal communication, 2026) that the affine base of Sections 7–9 can be replaced by a small-cancellation construction as in [FFF]. His finitely presented torsion-free property-(T) group  $P$  contains a copy of every finitely presented torsion-free group, in particular a direct product  $P_1 \times P_2$  with  $P_i \cong P$ . Then  $\alpha: P \xrightarrow{\cong} P_1 \leq P$ , together with any  $a \in P_2 \setminus \{1\}$ , satisfies the hypotheses of Theorem 5.1. The group  $E(P, \alpha, a)$  contains involutions whatever the base. The affine base is used for the explicit presentation of Theorem A and for Theorems D and E.

The following theorem is the explicit instance, developed in Sections 7–9. **Theorem A** (explicit Kazhdan–Clifford group). *Let  $E$  be the group on the eight generators  $v_1, v_2, v_3, x, y, z, t, c$  with the forty-one relators displayed in Definition 7.1. Put*

$$w = [tct^{-1}, v_1(tct^{-1})v_1^{-1}].$$

*The defining relations make  $w$  central; together with  $c^2 = 1$ , centrality implies  $w^2 = 1$ . Then:*

- (1)  $w \neq 1$  in  $E$ , so  $w$  is a central involution;
- (2) for every sequence  $(d_n)$  of positive integers and every group homomorphism  $\Theta: E \rightarrow U_{\text{cor}}((d_n))$  into the corresponding unitary norm corona (2), one has  $\Theta(w) = 1$ .

In particular  $E$  is not MF, and neither  $C_{\max}^*(E)$  nor  $C_{\text{red}}^*(E)$  is an MF  $C^*$ -algebra. [Lean: GroupApproximation.LiteralNonMFEndpoint.manuscriptTheoremA]

Theorem A has an equivalent uniform local-model form:

**Corollary 5.3** (uniform finite operator-norm obstruction). *There are a number  $\delta > 0$  and a finite set  $F_0 \subset E$  such that, in every matrix dimension, any unitary-valued map  $\phi$  satisfying*

$$\|\phi(gh) - \phi(g)\phi(h)\| \leq \delta \quad (g, h \in F_0)$$

*also satisfies  $\|\phi(w) - 1\| < 1$ .* [Lean: GroupApproximation.LiteralUniformObstruction.literal\_uniform\_operatorNorm\_obstruction]

*Proof.* Suppose no such pair  $(\delta, F_0)$  existed. For each  $n$ , choose a model violating the conclusion for  $\delta = 1/n$  and the  $n$ th set of an exhaustion of  $E$ . These models form an operator-norm asymptotic representation of  $E$  whose values at  $w$  stay at distance at least 1 from the identity. Its class in the unitary-sequence corona is a corona representation detecting  $w$ , contradicting Theorem A(2). The compactness argument is not quantitative; determining an effective modulus from the rational spectral certificate of Section 7 remains open.  $\square$

The same construction works for every exponent. For  $m \geq 2$ , let  $E_m$  be the same eight-generator presentation, replacing the three relations  $tv_it^{-1} = v_i^2$  by  $tv_it^{-1} = v_i^m$  and leaving every other relator unchanged; thus  $E_2 = E$ . Write  $w_m$  for the image of the same marked word  $w$ .

**Corollary 5.4** (the scaling family). *For every natural number  $m \geq 2$ , the group  $E_m$  is finitely presented and  $w_m$  is a nontrivial central involution mapped to the identity by every homomorphism from  $E_m$  to every unitary norm matrix corona. Consequently  $E_m$  is not MF, and neither  $C_{\max}^*(E_m)$  nor  $C_{\text{red}}^*(E_m)$  is an MF  $C^*$ -algebra.* [Lean: GroupApproximation.ScalingFamilyEndpoint.manuscriptTheoremFamily]

*Proof.* Replace the affine dilation  $\text{diag}(2, 2, 2, 1)$  by  $\text{diag}(m, m, m, 1)$  (both acting as in Lemma 8.2). It conjugates each integral translation  $v_i$  to  $v_i^m$  and fixes the three linear generators. Its inverse sends  $v_1$  to translation by  $e_1/m$ , which is not integral; hence the two Clifford sites remain distinct and  $w_m$  maps to  $-1$ . The Kazhdan base, the one-sided conjugation argument, and the central-sign criterion are independent of  $m$ .  $\square$

For the proof of Theorem A, put  $d = tct^{-1}$  and  $u = [d, v_1]$ . Lemma 9.1 gives the identity  $w = u^2$ , so  $w$  lies in the compression defect of the datum  $(\mathcal{B}, \iota, t, c)$ , and the central-sign criterion applies at the marked word of  $E$ . In concrete terms: for a corona representation that detects  $w$ , the negative corner of  $w$  gives a model in which  $w = -1$  exactly, while Kazhdan transport gives  $u \sim_2 1$  and hence  $w = u^2 \sim_2 1$ . This contradiction is independently formalized. [Lean: GroupApproximation.LiteralNonMFEndpoint.negativeCorner\_kazhdanTransport\_contradiction]

Nontriviality of  $w$  is witnessed by an explicit representation of  $E$  into a semidirect product of a Clifford group by a mapping-telescope group built from the exact rational affine matrix model (Section 8).

**Consequences** For  $E$ , the datum is  $(\mathcal{B}, \iota, t, c)$  and the relevant finite normal subgroup of the defect is  $\{1, w\}$ . Corollary 11.8 gives the resulting obstruction for  $C^*$ -subalgebras of norm matrix coronas, and Theorem C shows that finite-dimensional collapse by itself does not decide whether the marked word is trivial. The group  $E$  also yields a finite universal Horn sentence valid in every MF group and falsified by  $E$ , hence a nonempty clopen set of non-MF marked groups (Proposition 11.21 and Corollary 11.22), and the failures of quotient and extension permanence (Corollary 11.12 and Section 11).

## 6 THE FINITE-DIMENSIONAL OBSTRUCTION

The finite-dimensional analogue of the obstruction is elementary: it needs finite dimensionality and the one-sided conjugation and centralizer relations alone, and its conclusion covers the entire intrinsic defect (4):

**Theorem B** (the exact finite-dimensional analogue). *Let  $k$  be a field,  $V$  a finite-dimensional  $k$ -vector space,  $H$  a group,  $\Gamma \leq H$  a subgroup, and  $t, c \in H$ ,  $a \in \Gamma$  elements with  $t\Gamma t^{-1} \subseteq \Gamma$  and  $c\gamma = \gamma c$  for all  $\gamma \in \Gamma$ . Then every homomorphism  $\pi: H \rightarrow \text{GL}(V)$  satisfies*

$$\pi([tct^{-1}, a(tct^{-1})a^{-1}]) = 1.$$

*More generally, every such  $\pi$  sends the entire subgroup  $\mathfrak{D}(H, \Gamma)$  to the identity.*

[Lean: GroupApproximation.ManuscriptExactWrappers.manuscriptTheoremB]

Applied to  $H = E$ , Theorem B shows that *every* finite-dimensional representation of  $E$  over *every* field sends  $w$  to the identity. Consequently finite-dimensional linear representations do not separate the points of  $E$ , no finite quotient can witness  $w \neq 1$ , and the infinite Clifford witness of Section 8 cannot be replaced by a finite-dimensional one. In the terminology of [BDL, Definition 1.2], the central involution  $w$ , invisible in every finite quotient, makes  $E$  a group of *Deligne type*. Such groups were proposed there as the natural candidate counterexamples for all three remaining approximation problems, and  $E$  realizes that proposal at the operator-norm endpoint. Property (T) enters the corona argument through the Kazhdan fixed-space projection of the adjoint actions — in quantitative form, a finite-stage spectral subspace at each coordinate, where the equal-rank reversal estimate of Appendix B plays the role that equal dimension plays in the exact argument below.

*Proof of Theorem B.* Let  $\pi: H \rightarrow \text{GL}(V)$  be a homomorphism and let

$$C = \{x \in \text{End}_k(V) : \pi(\gamma)x = x\pi(\gamma) \text{ for all } \gamma \in \Gamma\}$$

be the commutant of  $\pi(\Gamma)$ , a linear subspace of  $\text{End}_k(V)$ . Let  $u = \pi(t)$  and consider the linear automorphism  $\Phi(x) = uxu^{-1}$  of  $\text{End}_k(V)$ .

We claim  $C \subseteq \Phi(C)$ . Let  $x \in C$  and set  $y = u^{-1}xu$ , so that  $x = \Phi(y)$ . For  $\gamma \in \Gamma$  we have  $t\gamma t^{-1} \in \Gamma$ , hence  $\pi(t\gamma t^{-1})$  commutes with  $x$ , and therefore

$$\pi(\gamma)y = u^{-1}\pi(t\gamma t^{-1})xu = u^{-1}x\pi(t\gamma t^{-1})u = y\pi(\gamma).$$

Thus  $y \in C$ , proving the claim. Since  $\Phi$  is a linear isomorphism,  $\dim_k \Phi(C) = \dim_k C < \infty$ , and the inclusion  $C \subseteq \Phi(C)$  forces

$$(7) \quad C = \Phi(C) = uCu^{-1}.$$

Now  $\pi(c) \in C$  because  $c$  centralizes  $\Gamma$ . By (7),  $\pi(tct^{-1}) = u\pi(c)u^{-1} \in C$ . Since  $a \in \Gamma$ , the element  $\pi(tct^{-1})$  commutes with  $\pi(a)$ , so  $\pi(a(tct^{-1})a^{-1}) = \pi(tct^{-1})$ , and therefore

$$\pi([tct^{-1}, a(tct^{-1})a^{-1}]) = [\pi(tct^{-1}), \pi(tct^{-1})] = 1. \quad \square$$

The same calculation proves the final, subgroup-valued assertion of Theorem B. The elements of  $H$  whose conjugation action preserves  $C$  in both directions form a subgroup. The argument above puts every one-sided conjugator in that subgroup, and therefore puts all of  $G_{\text{comp}}(\Gamma)$  in it. Thus, for  $g \in G_{\text{comp}}(\Gamma)$  and  $z \in C_H(\Gamma)$ , the operator  $\pi(gzg^{-1})$  still belongs to  $C$  and commutes with every  $\pi(\ell)$ ,  $\ell \in \Gamma$ . Hence every displayed generator in (4) lies in  $\ker \pi$ ; since  $\ker \pi \trianglelefteq H$ , the whole normal closure  $\mathfrak{D}(H, \Gamma)$  does as well.

**Corollary 6.1.** *The group  $E$  of Definition 7.1 admits no injective homomorphism into  $\text{GL}(V)$  for any finite-dimensional vector space  $V$  over any field. More strongly, every finite-dimensional linear representation maps  $w$  to the identity, and every homomorphism to a finite group maps  $w$  to the identity.* [Lean: GroupApproximation.LiteralFiniteDimensionalObstruction.manuscriptFiniteDimensionalConsequences]

*Proof.* Apply Theorem B with  $H = E$ , the image of  $\mathcal{B}$ , and the elements  $t, c, v_1$ ; the hypotheses hold by the defining relations (10). Every finite-dimensional representation kills  $w$ , and  $w \neq 1$  by Proposition 8.3. For a finite target, compose with its faithful left regular representation on the group algebra over  $\mathbb{Q}$ . The composite is a finite-dimensional linear representation, so it kills  $w$ . Evaluating the resulting identity matrix at the basis vector of the identity element returns the image of  $w$  in the finite group.  $\square$

*Remark 6.2* (the invariant-size principle). The proof of Theorem B used exactly one feature of finite dimensionality: a subspace of a finite-dimensional space that contains an isomorphic copy of itself equals it. In abstract form: suppose a conjugation-invariant size  $\sigma$  with values in  $\mathbb{N}$  is defined on the subgroups of a group  $Q$  and satisfies the separation property that  $K \leq L$  with  $\sigma(L) \leq \sigma(K)$  forces  $K = L$ . If  $gKg^{-1} \subseteq K$ , then  $gKg^{-1}$  and  $K$  have equal size, so the inclusion is an equality: such a target admits no strict one-sided compression. [Lean: GroupApproximation.ManuscriptExactWrappers.manuscriptInvariantSizePrinciple] Cardinality on the subgroups of a finite group and the dimension of the fixed space of the adjoint action in a finite-dimensional representation both qualify: finite quotients and finite-dimensional linear

representations are blind to strict one-sided compression for one and the same reason. The corona argument of Theorem A is the analytic counterpart of this principle at infinite dimension.

*Remark 6.3* (chain conditions suffice). The separation property of the size can be dropped, at the price of concluding only that the invariant is blind. An order isomorphism of a poset with no infinite strictly ascending chain fixes every point that it moves weakly upward, and dually with descending chains; consequently a conjugation-equivariant monotone invariant with chain-condition values takes the same value on  $\Gamma$  and on  $t\Gamma t^{-1}$ . In particular, for every finite-dimensional representation  $\sigma$  of  $H$  over every field, the fixed subspaces of the subgroup and of its one-sided conjugate coincide. [Lean: GroupApproximation.manuscriptChainConditionRigidity] Applied to the representations induced on tensor constructions —  $V^{\otimes p} \otimes \bar{V}^{\otimes q}$ , finite direct sums, duals, Hom-spaces — compression creates no invariant tensors of any finite type; the commutant identity (7) is the case  $\text{End}(V) \cong V \otimes \bar{V}$ . Taking the invariant to be the Zariski closure of the image in  $\text{GL}(V)$ , whose closed subgroups satisfy the descending chain condition, gives  $\overline{\sigma(t\Gamma t^{-1})}^Z = \overline{\sigma(\Gamma)}^Z$ : an exact finite-dimensional representation assigns the compressed subgroup and the subgroup the same algebraic envelope. The chain condition for algebraic subgroups is classical and outside the formal development. Theorem 3.4 is the asymptotic counterpart at fixed tensor type.

Why the corona argument is different. An MF model is only asymptotically multiplicative, so its coordinate fixed spaces are not literal invariant subspaces. Operator-norm control keeps the coordinate adjoint actions almost multiplicative on the Hilbert–Schmidt spaces. Property (T) then supplies an almost-fixed spectral subspace. The equal-rank reversal estimate of Appendix B turns its one-sided compression into equality. These are the quantitative counterparts of the exact finite-dimensional steps above.

## 7 THE EXPLICIT PRESENTATION AND ITS PROPERTY-(T) BASE

We work throughout with the groups defined by the relators below. Classically the base is the affine group  $\mathbb{Z}^3 \rtimes \text{SL}_3(\mathbb{Z})$ , a familiar Kazhdan group (Remark 7.3). The proof of property (T) given here is nevertheless intrinsic to the displayed presentations and independent of the identification: an exact rational spectral certificate for an auxiliary Steinberg-type presentation, an explicit surjection onto the linear presentation, and a fixed-point bridge from the linear subgroup to the full base. First define the *linear presentation*

$$(8) \quad \mathcal{R} = \left\langle x, y, z \mid \begin{array}{l} x^3 = y^3 = z^2 = (xz)^3 = (yz)^3 = 1, \\ (x^{-1}zxy)^2 = (y^{-1}zyx)^2 = (xy)^6 = 1 \end{array} \right\rangle,$$

and the *base*

(9)

$$\mathcal{B} = \left\langle v_1, v_2, v_3, x, y, z \left| \begin{array}{l} \text{the eight relations in (8),} \\ [v_1, v_2] = [v_1, v_3] = [v_2, v_3] = 1, \\ xv_1x^{-1} = v_3, \quad xv_2x^{-1} = v_1, \quad xv_3x^{-1} = v_2, \\ yv_1y^{-1} = v_1, \quad yv_2y^{-1} = v_2^{-1}v_3, \quad yv_3y^{-1} = v_1v_2^{-1}, \\ zv_1z^{-1} = v_2v_3^{-1}, \quad zv_2z^{-1} = v_1v_3^{-1}, \quad zv_3z^{-1} = v_3^{-1} \end{array} \right. \right\rangle.$$

Write  $\iota: \mathcal{B} \rightarrow E$  for the homomorphism induced by the first six generators of the presentation below. This map is not assumed to be injective.

**Definition 7.1.** Use the six base generators and the relations displayed in (9). Define

$$(10) \quad E = \left\langle v_1, v_2, v_3, x, y, z, t, c \left| \begin{array}{l} \text{all relations in (9),} \\ tv_it^{-1} = v_i^2 \quad (1 \leq i \leq 3), \\ txt^{-1} = x, \quad tyt^{-1} = y, \quad tzt^{-1} = z, \\ c^2 = 1, \\ [c, g] = 1 \quad (g \in \{v_1, v_2, v_3, x, y, z\}), \\ [w, g] = 1 \quad (g \in \{v_1, v_2, v_3, x, y, z, t, c\}) \end{array} \right. \right\rangle,$$

where  $w$  abbreviates the word  $[tct^{-1}, v_1(tct^{-1})v_1^{-1}]$ . This is an explicit finite presentation with eight generators and forty-one relators: the twenty base relators, six stable-letter relators,  $c^2$ , the six relations making  $c$  commute with the base generators, and eight centrality relators. Its defining relations make  $w$  central, and centrality together with  $c^2 = 1$  implies  $w^2 = 1$  by (6). The relations on the six base generators also give

$$t\iota(\mathcal{B})t^{-1} \subseteq \iota(\mathcal{B}), \quad [c, \iota(\mathcal{B})] = 1.$$

[Lean: GroupApproximation.LiteralNonMFPresentation.manuscriptLiteralPresentation]

Six generators suffice. The eight-generator presentation is used because it displays the structure the proof runs on: the three commuting translations, the linear generators acting on them, the stable letter, and the lamp. The group  $E$  is nevertheless generated by the six elements  $v_1, x, y, z, t, c$ . [Lean: GroupApproximation.LiteralSixGenerator.literal\_sixGenerated\_finitelyPresented\_nonMF]

There is also a Tietze-equivalent presentation on these six generators with thirty-two relators. In (10), eliminate  $v_3$  and  $v_2$  by the Tietze substitutions

$$v_3 = xv_1x^{-1}, \quad v_2 = x^2v_1x^{-2}.$$

Delete the two relations used as definitions. After substitution, the remaining  $x$ -action relation follows from  $x^3 = 1$ . The two stable-letter relations for  $v_2, v_3$  follow from the one for  $v_1$  and  $[t, x] = 1$ ; the two  $c$ -commutation relations and two centrality relations for  $v_2, v_3$  similarly follow from those for  $v_1$  and  $x$ . These six redundant relators may therefore be deleted. Starting from



forty-one relators, the two Tietze definitions, the remaining  $x$ -action relation, and those six consequences leave thirty-two, on the generators  $v_1, x, y, z, t, c$ .

Figure 1 isolates the geometry used later in the analytic argument. The element  $d = tct^{-1}$  centralizes the compressed copy, not necessarily all of  $\iota(\mathcal{B})$ . The marked commutator measures precisely that failure after conjugating  $d$  by  $\iota(v_1)$ .

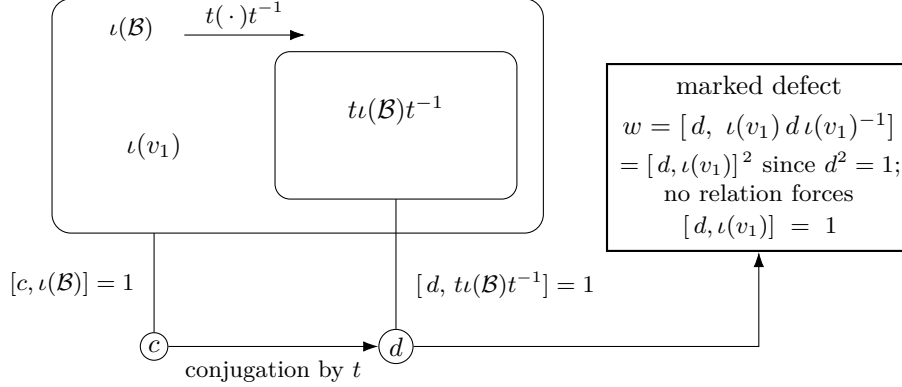


Figure 1: The one-sided conjugation configuration. Conjugation by  $t$  carries the image of the base into itself. It sends  $c$  to  $d = tct^{-1}$ : the relations force  $c$  to commute with all of  $\iota(\mathcal{B})$  and force  $d$  to commute with the compressed copy, but do not require  $d$  to commute with  $\iota(v_1)$ . The marked word  $w$  is that commutation defect. Only the one-sided inclusion is used by the analytic criterion.

The conjugation argument uses only the one-sided inclusion and centralizer relations shown in Figure 1. Injectivity of the base map is established independently by the Clifford realization.

### 7.1 Property (T) of the base

**Proposition 7.2.** *The twenty-relator group  $\mathcal{B}$  has property (T), in both the real-orthogonal and complex-unitary formulations.* [Lean: GroupApproximation.L iterBaseP13PropertyTBridge.manuscriptBaseHasKazhdanPropertyT]

*Proof. Step 1: a certified Steinberg-type presentation.* Let  $P_{13}$  be the group on the six generators  $e_{12}, e_{13}, e_{21}, e_{23}, e_{31}, e_{32}$  with the thirteen relators

$$\begin{aligned} [e_{12}, e_{23}] &= e_{13}, & [e_{13}, e_{32}] &= e_{12}, & [e_{21}, e_{13}] &= e_{23}, \\ [e_{23}, e_{31}] &= e_{21}, & [e_{31}, e_{12}] &= e_{32}, & [e_{32}, e_{21}] &= e_{31}, \\ [e_{12}, e_{13}] &= [e_{12}, e_{32}] = [e_{13}, e_{23}] = [e_{21}, e_{23}] = [e_{21}, e_{31}] = [e_{31}, e_{32}] = 1, \\ (e_{12} e_{21}^{-1} e_{12})^4 &= 1, \end{aligned}$$

the elementary-matrix relations in rank three. An exact sum-of-squares certificate with integer matrix data, verified by denominator-free rational

arithmetic, proves the degree-one Hodge estimate for  $P_{13}$  with spectral gap  $\frac{1}{500}$ . Testing it on coboundaries gives, in every orthogonal representation, the quadratic gap

$$\|\Delta\xi\|^2 \geq \frac{1}{500} \langle \Delta\xi, \xi \rangle$$

for the six-generator Laplacian  $\Delta$ . The gap places the spectrum of  $\Delta$  in  $\{0\} \cup [\frac{1}{500}, \infty)$ . The kernel of  $\Delta$  is exactly the space of invariant vectors. Hence a unit vector almost fixed by the six generators has small Laplacian energy and is close to an invariant vector. This yields a Kazhdan pair:  $P_{13}$  has property (T). Certificates of this kind go back to Ozawa's algebraic characterization of property (T) [Oz] and its computer-assisted implementations for  $\mathrm{SL}_3(\mathbb{Z})$  [NT].

*Step 2: the explicit surjection onto the linear presentation.* Map the six generators to the words

$$\begin{aligned} e_{12} &\mapsto y^{-1}xyz^{-1}x^{-1}, & e_{13} &\mapsto xzy^{-1}x^{-1}y, & e_{21} &\mapsto y^{-1}xyz, \\ e_{23} &\mapsto x^{-1}y^{-1}xzy^{-1}, & e_{31} &\mapsto xy^{-1}xzy^{-1}x, & e_{32} &\mapsto x^{-1}zy^{-1}x^{-1}y^{-1}x^{-1} \end{aligned}$$

in the linear presentation  $\mathcal{R}$  of (8). A word-by-word verification inside  $\mathcal{R}$ , using only its eight displayed relators, checks that all thirteen relators of  $P_{13}$  map to the identity, so the assignment defines a homomorphism  $P_{13} \rightarrow \mathcal{R}$ . The same computation exhibits explicit words in the six images whose values are  $z$  and  $xy$ . A further computation shows that  $z$  and  $xy$  generate  $\mathcal{R}$ . Hence the homomorphism is surjective. Property (T) passes to quotients, so  $\mathcal{R}$  has property (T).

*Step 3: the fixed-point bridge to the base.* The generators  $x, y, z$  of  $\mathcal{B}$  satisfy the relators of  $\mathcal{R}$ , so  $\mathcal{R}$  maps to the linear subgroup of  $\mathcal{B}$  they generate, and Step 2 gives a finite set  $S$  of words in  $x, y, z$  and a  $\kappa > 0$  forming a Kazhdan pair for it. Let  $\rho$  be an orthogonal representation of  $\mathcal{B}$  with a unit vector  $\xi$  displaced by less than  $\kappa/64$  by every element of the finite control set consisting of  $S$  together with the translations  $v_2, v_3$ . Three estimates produce an invariant vector near  $\xi$ .

- (1) Projecting  $\xi$  to the subspace of vectors fixed by the linear subgroup gives such a fixed vector  $p$  with  $\|p - \xi\| < 1/64$ , by the moving-projection estimate for the Kazhdan pair.
- (2) Translations move  $p$  by a bound independent of the lattice vector. Three explicit words in  $x, y, z$  act on the translation lattice as elementary transvections. Consequently every lattice element is a product of two linear-subgroup conjugates of basis translations — a two-conjugate normal form proved inside the presentation. A conjugate  $rvr^{-1}$  with  $r$  in the linear subgroup displaces the fixed vector  $p$  exactly as much as  $v$  does. Each control translation displaces  $p$  by less than  $3/64$ , because its displacement bound at  $\xi$  transfers across  $\|p - \xi\|$ . Hence every translation displaces  $p$  by at most  $1/8$ .
- (3) The orbit of  $p$  under the translation subgroup is bounded, so the Hilbert-space circumcenter argument applies: the projection  $\eta$  of  $p$

to the translation-fixed subspace satisfies  $\|\eta - p\| \leq 1/8$ . The linear subgroup normalizes the translation subgroup and fixes  $p$ , so  $\eta$  is fixed by it as well. Translations and the linear generators generate  $\mathcal{B}$ , so  $\eta$  is  $\mathcal{B}$ -invariant. Finally,  $\|\eta - \xi\| \leq 1/8 + 1/64 < 1$  keeps  $\eta$  nonzero.

Thus almost-invariant vectors produce invariant vectors and  $\mathcal{B}$  has property (T) in the real-orthogonal formulation; realification and complexification identify it with the complex-unitary formulation.  $\square$

*Remark 7.3* (the classical identification). Classically, the eight relations (8) present  $\mathrm{SL}_3(\mathbb{Z})$  [CRW, Theorem 2]; the matrix-labelled form of the generators in Lemma 8.2 is also recorded in [CLV]. The base  $\mathcal{B}$  is then the affine group  $\mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$ . Property (T) of that affine group is classical [BHV, Example 1.7.4(i)]. Thus property (T) of  $\mathcal{B}$  also follows from the classical results; the proof above is independent of them.

## 8 THE WITNESS: $w \neq 1$

Clifford signs, shifts, and HNN doubling have appeared together before in approximate-representation theory, notably in Slofstra's hyperlinear-profile construction [Slo18, Section 2]. The construction below uses the Clifford lamp as an explicit algebraic witness.

**Construction 8.1** (Clifford lamp groups). For a set  $X$ , use the real Clifford-algebra convention generated by  $(c_x)_{x \in X}$  with  $c_x^2 = 1$  and  $c_x c_y = -c_y c_x$  for  $x \neq y$ , and let  $\mathrm{CLamp}(X)$  be the subgroup of units generated by  $-1$  and the  $c_x$ . Write  $\zeta = -1$ . In this concrete group,

$$\zeta^2 = c_x^2 = 1, \quad c_x c_y = \zeta c_y c_x \quad (x \neq y).$$

The sign  $\zeta$  is nontrivial. Thus  $\mathrm{CLamp}(X)$  is a central extension, by the sign  $\zeta$ , of the elementary abelian 2-group on the sites  $X$ : distinct site involutions commute only up to  $\zeta$ , unlike the commuting lamps of a classical lamplighter. Every permutation of  $X$  acts on this group by  $c_x \mapsto c_{\sigma(x)}$  and fixes  $\zeta$ . [Lean: GroupApproximation.ManuscriptExactWrappers.manuscriptCliffordConstruction]

**Lemma 8.2** (linear model). *For  $M \in \mathrm{GL}_3(\mathbb{Q})$  and  $u \in \mathbb{Q}^3$ , write*

$$A(M, u) = \begin{pmatrix} M & u \\ 0 & 1 \end{pmatrix}.$$

*Assign  $v_i \mapsto A(I_3, e_i)$  and*

$$\begin{aligned} x &\mapsto A\left(\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}, 0\right), & y &\mapsto A\left(\begin{pmatrix} 1 & 0 & 1 \\ 0 & -1 & -1 \\ 0 & 1 & 0 \end{pmatrix}, 0\right), \\ z &\mapsto A\left(\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ -1 & -1 & -1 \end{pmatrix}, 0\right). \end{aligned}$$

Let  $\bar{\Gamma} \leq \mathrm{GL}_4(\mathbb{Q})$  be the subgroup generated by these six matrices, and put  $D = \mathrm{diag}(2, 2, 2, 1)$ . Direct matrix computation proves that the six matrices satisfy all twenty relators of  $\mathcal{B}$  and that

$$\bar{\alpha}(u) = DuD^{-1}$$

is an injective endomorphism of  $\bar{\Gamma}$  which squares the three translation generators and fixes  $x, y, z$ . Moreover the matrix assigned to  $v_1$  does not lie in  $\mathrm{range}(\bar{\alpha})$ . [Lean: GroupApproximation.ManuscriptExactWrappers.manuscriptLinearModel]

*Proof.* The twenty relator checks reduce to sixteen rational identities for each matrix equation. Conjugation by  $D$  gives the six stable-letter equations and is injective because it is conjugation by an invertible matrix. Every displayed generator and its explicitly computed inverse has integral entries; equivalently, all six generators lie in  $\mathrm{GL}_4(\mathbb{Z})$ . Hence  $\bar{\Gamma} \leq \mathrm{GL}_4(\mathbb{Z})$ . However  $D^{-1}\bar{v}_1D$  has entry  $1/2$  in position  $(1, 4)$ , so it is not in  $\bar{\Gamma}$ . This proves the last assertion.  $\square$

Form the direct limit  $T = \varinjlim(\bar{\Gamma}, \bar{\alpha})$ , the algebraic mapping telescope of  $\bar{\alpha}$ , extend it by its shift  $\tau$ , and call the resulting group  $V = T \rtimes \mathbb{Z}$ . Let  $j: \bar{\Gamma} \hookrightarrow V$  be the level-zero map. Put  $X = V/j(\bar{\Gamma})$ , with root  $o = j(\bar{\Gamma})$ . The final assertion of Lemma 8.2 states that

$$\tau o \neq j(\bar{v}_1)\tau o.$$

Thus the Clifford generators at these two sites anticommute.

**Proposition 8.3.**  $w \neq 1$  in  $E$ . [Lean: GroupApproximation.LiteralNonMFLLinearWitness.literal\_mark\_ne\_one]

*Proof.* Form  $W := \mathrm{Cllamp}(X) \rtimes V$  as in Construction 8.1. Define a map on the generators of (10) by

$$\begin{aligned} g &\mapsto j(\bar{g}) \in V & (g \in \{v_1, v_2, v_3, x, y, z\}), \\ t &\mapsto \tau, & c &\mapsto c_o. \end{aligned}$$

The exact checks in Lemma 8.2 give the twenty base and six stable-letter relations. The level-zero subgroup fixes  $o$ , so  $c_o$  commutes with every base generator, and  $c_o^2 = 1$ . Under this assignment,

$$tct^{-1} \mapsto c_{\tau o}, \quad v_1(tct^{-1})v_1^{-1} \mapsto c_{j(\bar{v}_1)\tau o},$$

and the two sites are distinct by Lemma 8.2, so

$$w \mapsto [c_{\tau o}, c_{j(\bar{v}_1)\tau o}] = -1 = \zeta.$$

Since  $\zeta$  is central in  $W$ , the remaining relations  $[w, g] = 1$  for  $g \in \{v_1, v_2, v_3, x, y, z, t, c\}$  are also satisfied. Hence the assignment extends to a group homomorphism  $E \rightarrow W$  sending  $w$  to  $\zeta \neq 1$ , and therefore  $w \neq 1$  in  $E$ .  $\square$

Scope of the witness. The target uses the infinite group  $\text{CLamp}(X)$  when  $X$  is infinite, but each group element involves only finitely many Clifford sites, and the remaining data are rational matrices. The mapping telescope supplies the required group, and the range characterization separates the two sites; no HNN normal-form theorem is needed.

## 9 THE DEFECT SQUARE AND THE PROOF OF THEOREM A

**Lemma 9.1** (the marked word is a defect square). *Let  $H_1$  be a group and  $d, a_1 \in H_1$  with  $d^2 = 1$ . Then*

$$[d, a_1 d a_1^{-1}] = [d, a_1]^2.$$

[Lean: GroupApproximation.commutator\_conjugate\_eq\_commutator\_sq\_of\_sq\_eq\_one]

*Proof.* Put  $b = a_1 d a_1^{-1}$ , so  $b^2 = a_1 d^2 a_1^{-1} = 1$ ; thus  $b^{-1} = b$  and  $d^{-1} = d$ . Then  $[d, a_1] = d a_1 d^{-1} a_1^{-1} = d (a_1 d a_1^{-1}) = db$ , and

$$[d, a_1]^2 = (db)^2 = db db = d b d^{-1} b^{-1} = [d, b]. \quad \square$$

*Proof of Theorem A.* Part (1) is Proposition 8.3. For part (2), let  $\Theta: E \rightarrow \text{U}_{\text{cor}}((d_n))$  be given and pass to the corona representation  $\kappa_{(d_n)} \circ \Theta: E \rightarrow \mathcal{U}(\mathcal{Q})$  of Lemma 2.2. Apply the central-sign criterion (Theorem 4.3) with  $\Gamma = \mathcal{B}$ , the canonical homomorphism  $\iota$ , the elements  $t, c \in E$ , and  $a = v_1$ . Its hypotheses are the displayed relators: Proposition 7.2 gives property (T) of  $\mathcal{B}$ ; the six stable-letter relators give  $t\iota(\mathcal{B})t^{-1} \subseteq \iota(\mathcal{B})$ ; and the six  $c$ -commutation relators give  $[c, \iota(\mathcal{B})] = 1$ . In  $E$  we have  $d^2 = (tct^{-1})^2 = tc^2t^{-1} = 1$ , so Lemma 9.1 with  $a_1 = \iota(v_1)$  identifies the marked word as the square of the conjugation defect  $u = [d, \iota(v_1)]$ :

$$w = [d, \iota(v_1) d \iota(v_1)^{-1}] = [d, \iota(v_1)]^2 = u^2.$$

Centrality of  $w$  is a defining relator; together with  $c^2 = 1$  it forces  $w^2 = 1$  by (6); and  $w \neq 1$  by part (1). The criterion therefore sends  $w$  to the identity under every corona representation of  $E$ . Since  $\kappa_{(d_n)}$  is injective,  $\Theta(w) = 1$ . This proves part (2).

For the final assertions of Theorem A: an MF embedding of  $E$  would be an injective corona representation, which is impossible because every corona representation sends the nontrivial element  $w$  to 1. Suppose  $C_{\text{red}}^*(E)$  were an MF algebra, with an injective, possibly nonunital  $*$ -homomorphism  $e: C_{\text{red}}^*(E) \rightarrow \mathcal{Q}$ . Complement correction gives the injective unitary homomorphism  $u \mapsto e(u) + (1 - e(1))$ . Composing it with the canonical map  $E \rightarrow \mathcal{U}(C_{\text{red}}^*(E))$ , and then with  $\kappa_{(d_n)}^{-1}$ , would exhibit an injective corona representation of  $E$ , because the map  $g \mapsto \lambda_g$  is injective. The canonical map  $E \rightarrow \mathcal{U}(C_{\text{max}}^*(E))$  is injective as well, because the left regular representation factors through  $C_{\text{max}}^*(E)$  and sends  $u_g$  to  $\lambda_g$ . Thus an MF embedding of  $C_{\text{max}}^*(E)$  would again produce an injective corona representation of  $E$  after the same  $\kappa_{(d_n)}^{-1}$  conversion.  $\square$

The marked radical element. The proof shows more than non-injectivity of each individual  $\Theta$ : the fixed element  $w$  lies in the kernel of *every* corona representation of  $E$ , so every corona representation factors through  $E/\langle w \rangle$  (note  $\langle w \rangle = \{1, w\}$  is central, hence normal). In the language of approximation radicals,  $w$  is a nontrivial element of the “MF radical” of  $E$ . This terminology is meant in analogy with Slofstra’s earlier quotients  $G^{\text{fin}}$  and  $G^{\text{fa}}$ , obtained by killing elements invisible to exact and approximate finite-dimensional representations, respectively [Slo17, Definitions 2.5–2.7]. By Theorem 4.2 the same holds in every countable group with a Kazhdan conjugation datum: every finite normal subgroup of its defect subgroup lies in the MF radical. The sign is thus a certificate of the obstruction rather than its source: the proof of Theorem 4.2 makes the entire compression defect of  $E$  — in particular the unsquared commutator  $u = [d, \iota(v_1)]$  — Hilbert–Schmidt invisible in every operator-norm almost representation, and  $w$  is the finite-order element that converts that invisibility into an exact operator-norm contradiction. Section 11 develops the radical; whether  $E/\langle w \rangle$  is MF — equivalently, whether the MF radical of  $E$  is exactly  $\{1, w\}$  — is open.

Corner ranks. The coordinate Reynolds corner in the proof of Theorem 4.2 discards zero-rank coordinates and equips each retained corner block with its own normalized trace, so all subsequent Hilbert–Schmidt estimates are relative to that block and independent of the ratio  $r_n/d_n$ . This is what makes the operator-norm setting insensitive to small relative ranks, which block the same argument in Hilbert–Schmidt metrics (Section 12).

## 10 THE CYCLIC COMPARISON

The finite-dimensional calculation of Section 6 does not by itself determine whether its marked word is trivial:

**Theorem C** (the cyclic comparison). *Define the marked presentation*

$$E_{\text{BS}} = \left\langle \gamma_0, t, c \left| \begin{array}{l} t\gamma_0 t^{-1} = \gamma_0^2, \quad c^2 = 1, \quad [c, \gamma_0] = 1, \\ w_{\text{BS}}^2 = 1, \quad [w_{\text{BS}}, \gamma_0] = [w_{\text{BS}}, t] = [w_{\text{BS}}, c] = 1 \end{array} \right. \right\rangle,$$

where  $w_{\text{BS}} = [tct^{-1}, \gamma_0(tct^{-1})\gamma_0^{-1}]$ . Then  $E_{\text{BS}}$  is finitely presented and  $w_{\text{BS}} \neq 1$ . There is an explicit surjective homomorphism from  $E_{\text{BS}}$  onto its realized Clifford quotient that maps  $w_{\text{BS}}$  to the nontrivial central sign. Nevertheless every finite-dimensional representation of  $E_{\text{BS}}$  over every field maps  $w_{\text{BS}}$  to the identity. [Lean: GroupApproximation.LiteralCyclicCalibration.manuscriptCyclicCalibration]

Thus exact finite-dimensional invisibility does not by itself force triviality of the marked word. The MF conclusion in Theorem A is the separate uniform assertion supplied by the fixed-space argument.

*Proof of Theorem C.* As with  $w$ , the relator  $w_{\text{BS}}^2 = 1$  is redundant:  $tct^{-1}$  and  $\gamma_0(tct^{-1})\gamma_0^{-1}$  are involutions, so centrality forces it by (6). It is retained only to display the marked pattern. Deleting it gives a six-relator presentation of

the same group. The stable relation gives  $t\langle\gamma_0\rangle t^{-1} \subseteq \langle\gamma_0\rangle$ , and  $c$  centralizes this cyclic base. The same commutant argument as in Theorem B therefore maps  $w_{\text{BS}}$  to the identity in every finite-dimensional representation over every field.

For nontriviality, use the affine realization of  $\text{BS}(1, 2)$  and its coset action. The root site is fixed by the cyclic base, while the two sites represented by  $t$  and  $\gamma_0 t$  are distinct. Attach Clifford generators to these sites and map  $c$  to the root lamp. All seven relators hold, and

$$w_{\text{BS}} \mapsto [c_{t\langle\gamma_0\rangle}, c_{\gamma_0 t\langle\gamma_0\rangle}] = -1.$$

Thus  $w_{\text{BS}} \neq 1$ . Restricting the target to the subgroup generated by the three displayed images gives the asserted surjective realized quotient.  $\square$

Sharpness of the Kazhdan hypothesis. The realized Clifford quotient in Theorem C is amenable, hence MF, and its quotient map detects  $w_{\text{BS}}$ . Thus some corona representation of  $E_{\text{BS}}$  separates the marked word from the identity, and property (T) cannot be omitted from the central-sign criterion. In the radical language of Section 11,  $w_{\text{BS}} \notin \text{Rad}_{\text{MF}}(E_{\text{BS}})$ . The affine group  $\text{BS}(1, 2) \cong \mathbb{Z}[1/2] \rtimes \mathbb{Z}$  is solvable. The Clifford lamp group on its coset space is locally finite, so their semidirect product is amenable. The realized quotient is a subgroup of this amenable group and is therefore amenable, hence MF by [TWW]. Its marked word is the nontrivial Clifford sign by the calculation above.

This detecting corona representation is necessarily approximate: neither the group  $E_{\text{BS}}$  nor its realized Clifford quotient is LEF, and the latter is not residually finite. Indeed, in every finite group the doubling relation makes the image of  $\langle\gamma_0^2\rangle$  equal to the image of  $\langle\gamma_0\rangle$ ; the lamp therefore commutes with the required “halved” element, and the marked word maps to the identity.

[Lean: GroupApproximation.CyclicBaseLEFObstruction.cyclicBase\_exactModel\_package]

## 11 CONSEQUENCES

This section develops the universal MF radical and the basic permanence facts, computes radicals by orbit collapse, constructs a sofic non-MF group, and proves the reduced-group- $C^*$  consequence. They all come from the same marked element.

### 11.1 The MF radical

**Definition 11.1.** For a countable group  $G_1$ , the *MF radical* is

$$\text{Rad}_{\text{MF}}(G_1) = \bigcap \{\ker \Theta : \Theta \text{ a corona representation of } G_1\},$$

a normal subgroup of  $G_1$ . This is the operator-norm analogue of the invisible subgroups of [Slo17, Definitions 2.5–2.7] and of the metric-ultraproduct approximation framework of [NST].

[Lean: GroupApproximation.ManuscriptExactWriters.manuscriptMFRadical]

**Lemma 11.2** (functoriality). *For every homomorphism  $\varphi: G_1 \rightarrow G_2$  of countable groups,*

$$\varphi(\text{Rad}_{\text{MF}}(G_1)) \subseteq \text{Rad}_{\text{MF}}(G_2) :$$

*the MF radical is functorial. In particular, if  $\varphi: E \rightarrow G_2$  satisfies  $\varphi(w) \neq 1$ , then  $G_2$  is not MF.* [Lean: GroupApproximation.ManuscriptExactWrappers.manuscriptRadicalPortability]

*Proof.* If  $x \in \text{Rad}_{\text{MF}}(G_1)$  and  $\Theta$  is a corona representation of  $G_2$ , then  $\Theta \circ \varphi$  is a corona representation of  $G_1$ , so  $\Theta(\varphi(x)) = 1$ . For the second statement,  $w \in \text{Rad}_{\text{MF}}(E)$  by Theorem A, so  $1 \neq \varphi(w) \in \text{Rad}_{\text{MF}}(G_2)$ , and an injective corona representation of  $G_2$  would kill  $\varphi(w)$ , a contradiction.  $\square$

**Proposition 11.3** (the universal MF quotient). *Let  $G_1$  be a countable group and  $R = \text{Rad}_{\text{MF}}(G_1)$ .*

- (1)  *$R$  is the kernel of a single corona representation of  $G_1$ .*
- (2)  *$G_1/R$  is MF.*
- (3) *Every homomorphism from  $G_1$  to an MF group factors through  $G_1/R$ .*

*In particular  $G_1$  is MF if and only if  $R = \{1\}$ .* [Lean: GroupApproximation.ManuscriptExactWrappers.manuscriptUniversalMFQuotient]

*Proof.* (3) If  $\varphi: G_1 \rightarrow M$  with  $M$  MF and  $j: M \rightarrow \text{U}_{\text{cor}}((d_n))$  is an injective corona representation, then  $j \circ \varphi$  is a corona representation of  $G_1$ , so  $j(\varphi(R)) = 1$  and, by injectivity of  $j$ ,  $\varphi(R) = 1$ .

(1) and (2). Work on  $Q = G_1/R$  from the outset. If  $1 \neq q \in Q$  and  $g \in G_1$  represents  $q$ , then  $g \notin R$ . Hence some corona representation of  $G_1$  detects  $g$ . Every corona representation kills  $R$  by definition, so this representation descends to one of  $Q$  that detects  $q$ . Thus the intersection of the kernels of all corona representations of the countable group  $Q$  is trivial.

Enumerate the nonidentity elements of  $Q$  and choose a detecting corona representation for each. Pass to coordinate unitary lifts, so that each detector becomes a sequential almost representation with its marked element separated from 1. Diagonalize at the coordinate level. At stage  $k$ , select in each of the first  $k$  detectors a coordinate at which multiplicativity holds within  $1/k$  on the first  $k$  group elements while the marked separation persists. Take the block sum of the  $k$  selected matrices. Operator norms of block sums are maxima, so the resulting single sequence is an almost representation detecting every nonidentity element. Its class in the unitary-sequence quotient, composed with the polar-correction isomorphism of Lemma 2.2, is a faithful corona representation of  $Q$ . This proves (2). Composing it with  $G_1 \twoheadrightarrow Q$  gives a single corona representation of  $G_1$  with kernel exactly  $R$ , proving (1). For the final equivalence: if  $G_1$  is MF, an injective corona representation forces  $R = \{1\}$ ; if  $R = \{1\}$ , then (2) says  $G_1 = G_1/R$  is MF.  $\square$

**Corollary 11.4** (exact radical from a candidate quotient). *Let  $N \trianglelefteq G_1$  satisfy  $N \leq \text{Rad}_{\text{MF}}(G_1)$ . Every corona representation of  $G_1$  factors uniquely*



through  $G_1/N$ . If in addition  $G_1/N$  is MF, then

$$\text{Rad}_{\text{MF}}(G_1) = N.$$

[Lean: GroupApproximation.ManuscriptExactWrappers.manuscriptExactRadicalFromCandidateQuotient]

*Proof.* The containment  $N \leq \ker \Theta$  gives the unique quotient factorization for every corona representation  $\Theta$ . For the reverse containment in the displayed equality, compose the quotient map  $G_1 \rightarrow G_1/N$  with an injective corona representation of  $G_1/N$ : its kernel is exactly  $N$ , while the MF radical lies in the kernel of every corona representation.  $\square$

The MF hypothesis on the quotient can be dropped entirely; what remains is an unconditional reduction.

**Corollary 11.5** (radical reduction to the quotient). *Let  $N \trianglelefteq G_1$  satisfy  $N \leq \text{Rad}_{\text{MF}}(G_1)$  and let  $q: G_1 \rightarrow G_1/N$  be the quotient map. Then*

$$\text{Rad}_{\text{MF}}(G_1) = q^{-1}(\text{Rad}_{\text{MF}}(G_1/N)).$$

[Lean: GroupApproximation.ManuscriptExactWrappers.manuscriptRadicalReductionToQuotient]

*Proof.* For the forward containment, Lemma 11.2 gives  $q(\text{Rad}_{\text{MF}}(G_1)) \subseteq \text{Rad}_{\text{MF}}(G_1/N)$ . Conversely, suppose  $q(x) \in \text{Rad}_{\text{MF}}(G_1/N)$  and let  $\Theta$  be a corona representation of  $G_1$ . Every corona representation of  $G_1$  kills  $N$  by hypothesis, so  $\Theta$  factors as  $\bar{\Theta} \circ q$  with  $\bar{\Theta}$  a corona representation of  $G_1/N$ ; hence  $\Theta(x) = \bar{\Theta}(q(x)) = 1$ .  $\square$

Corollary 11.4 is the case of trivial quotient radical. Corollary 11.5 needs no hypothesis on the quotient: the corona representation theory of  $G_1$  is that of  $G_1/N$ , whether or not the latter is MF.

By Corollary 11.5, the passage from  $N$  to the full preimage of the quotient radical is a closure operation on the normal subgroups of a fixed group.

**Proposition 11.6** (the MF closure of relations). *For a countable group  $G_1$  and  $N \trianglelefteq G_1$  with quotient map  $q: G_1 \rightarrow G_1/N$ , put*

$$\text{cl}_{\text{MF}}(N) = q^{-1}(\text{Rad}_{\text{MF}}(G_1/N)).$$

*An element belongs to  $\text{cl}_{\text{MF}}(N)$  exactly when every corona representation of  $G_1$  that kills  $N$  kills it.* [Lean: GroupApproximation.mem\_actualCoronaMFClosure\_iff]

*The operation  $N \mapsto \text{cl}_{\text{MF}}(N)$  is extensive, monotone, and idempotent on normal subgroups;  $G_1/\text{cl}_{\text{MF}}(N)$  is MF; and  $\text{cl}_{\text{MF}}(N) = N$  exactly when  $G_1/N$  is MF.* [Lean: GroupApproximation.manuscriptRelationClosure]

*Proof.* Corona representations of  $G_1$  killing  $N$  correspond to corona representations of  $G_1/N$ , which gives the membership characterization; extensivity, monotonicity, and idempotence are read off from it. The quotient clause is Proposition 11.3(2) transported along the third isomorphism theorem, and the fixed-point clause is the final equivalence of Proposition 11.3.  $\square$

For a presentation  $G = \langle X \mid R \rangle = F/N$ , the closure of  $N$  consists of the relations satisfied by every solution of  $R$  in every MF group, and the group is MF exactly when  $N$  is closed. Proposition 11.21 displays one finite fragment of this closure, and the marked element of Theorem A is an element of  $\text{cl}_{\text{MF}}(N) \setminus N$  for the forty-one relators.

**Proposition 11.7** (iterated sound rules). *Consider the normal subgroups of  $G_1$  reachable from the trivial subgroup by two operations: passing from  $N$  to any subgroup of  $\text{cl}_{\text{MF}}(N)$ , and taking unions of families already reached. Every subgroup so reached lies in  $\text{Rad}_{\text{MF}}(G_1)$ ; a reached subgroup with MF quotient equals  $\text{Rad}_{\text{MF}}(G_1)$ ; and if the whole group is reached, every homomorphism from  $G_1$  to a countable MF group is trivial.* [Lean: GroupApproximation.manuscriptSoundIterationClosure]

*Proof.* For  $N \leq \text{Rad}_{\text{MF}}(G_1)$ , Corollary 11.5 gives  $\text{cl}_{\text{MF}}(N) = \text{Rad}_{\text{MF}}(G_1)$ , so the first operation stays inside the radical, and unions of subgroups of the radical lie in the radical. The final two assertions are Corollary 11.4 and Proposition 11.3.  $\square$

Theorems 4.2, 4.6, and 11.14 are sound rules in this sense: each places a subgroup inside the radical of every countable group carrying its data, hence inside the radical of every reached quotient. Theorem 11.18 is the case in which a single application reaches the whole group. A rule instance can hold at a later stage without holding at an earlier one: modulo the earlier stage an element may acquire finite order, an orbit may begin to commute, or a subgroup may become normal.

**Corollary 11.8.** *Let  $A$  be a unital  $C^*$ -algebra admitting an injective, possibly nonunital,  $*$ -homomorphism into a norm matrix corona. Then the group  $E$  admits no injective homomorphism into  $\mathcal{U}(A)$ .* [Lean: GroupApproximation.LiteralNonMFConsequences.literal\_no\_faithful\_corona\_subalgebra\_target]

*Proof.* Let  $j: A \rightarrow Q$  be the possibly nonunital inclusion into the ambient corona and let  $\phi: E \rightarrow \mathcal{U}(A)$ . Complement correction gives

$$\hat{j}(u) = j(u) + 1 - j(1_A) \in \mathcal{U}(Q).$$

This is an injective homomorphism on  $\mathcal{U}(A)$ , so  $\hat{j} \circ \phi$  is a corona representation of  $E$ . By Theorem A(2),  $(\hat{j} \circ \phi)(w) = 1$ ; injectivity of  $\hat{j}$  gives  $\phi(w) = 1$ . Proposition 8.3 has  $w \neq 1$ , so  $\phi$  cannot be injective.  $\square$

## 11.2 Permanence and its failure

**Lemma 11.9** (subgroups). *Every subgroup of an MF group is MF.* [Lean: GroupApproximation.IsOperatorMF.subgroup]

*Proof.* Restrict an injective corona representation to the subgroup: injectivity and the homomorphism law are preserved.  $\square$

**Lemma 11.10** (residually finite groups). *Every countable residually finite group is MF.* [Lean: GroupApproximation.isOperatorMF\_of\_residuallyFinite]

*Proof.* Residual finiteness makes the group LEF. For each finite test set  $F$ , choose a finite quotient injective on  $F$ . Composing with its left regular permutation action gives an *exact* finite local model: every product of elements of  $F$  is represented with zero multiplicative defect. Distinct elements of  $F$  map to distinct permutations; at a point where the two permutations disagree, the difference of the two permutation matrices has a column of norm at least 1, so the model separates distinct elements of  $F$  by operator norm at least 1. These exact permutation models are operator-norm local models with separation constant 1, so the group is MF by the local-to-corona direction of Proposition 1.1.  $\square$

**Lemma 11.11** (locally finite groups). *Every countable locally finite group is MF.* [Lean: GroupApproximation.isOperatorMF\_of\_locallyFinite]

*Proof.* Given a finite test set  $F$ , local finiteness supplies one finite subgroup  $K$  containing it. Choose any map from the group to  $K$  that restricts to the identity on the test set. Applying the left regular permutation action of  $K$  to it gives an exact finite local model on  $F$ , separating distinct elements by operator norm at least 1 exactly as in Lemma 11.10. Thus the group is LEF, and it is MF by the local-to-corona direction of Proposition 1.1.  $\square$

**Corollary 11.12** (quotients). *MF groups are not closed under quotients: the free group  $F_8$  is MF, and the explicit group  $E$  is its quotient but is not MF.* [Lean: GroupApproximation.LiteralMFQuotientControls.manuscriptQuotientNonclosure]

*Proof.* Map the eight free generators onto the eight displayed generators of  $E$ . The resulting homomorphism is surjective because those generators define the displayed presentation. The group  $F_8$  is residually finite, hence MF by Lemma 11.10. Theorem A shows that  $E$  is not MF.  $\square$

**11.3 Orbit collapse and defect saturation** The criteria of Section 4 bound the radical from below by torsion or rigidity already present in the defect subgroup. This subsection gives a second application of one-sided compression, under a commuting-orbit hypothesis in place of centrality: every corona representation identifies an entire conjugation orbit of an involution, so whole families of commutators — not one marked element — enter the radical, and by Corollary 11.5 the radical computation reduces to the quotient by all of them.

**Definition 11.13** (involutive compression witness). Let  $H$  be a group,  $L \leq H$  a subgroup, and  $s \in H$ . An element  $k \in H$  is an *involutive compression witness* for  $(L, s)$  if

- (W1)  $k^2 = 1$ ;
- (W2)  $s\gamma s^{-1}$  commutes with  $k$  for every  $\gamma \in L$ ;
- (W3) the conjugates  $\gamma_1 k \gamma_1^{-1}$  and  $\gamma_2 k \gamma_2^{-1}$  commute for all  $\gamma_1, \gamma_2 \in L$ .

The *involutive collapse defect* of the pair  $(L, s)$  is the normal subgroup

$$D_{\text{coll}}(L, s) = \langle\langle [\gamma, k] : \gamma \in L, k \text{ a witness for } (L, s) \rangle\rangle_H.$$

[Lean: GroupApproximation.ManuscriptExactWrappers.manuscriptInvolutiveCollapsePattern]

In a permutational wreath product  $K^{(X)} \rtimes G$  with  $L \leq G$  compressed by  $s$  and  $x_0 \in X$  a point fixed by  $sLs^{-1}$ , every involution of the lamp copy at  $x_0$  is a witness: lamps at distinct sites commute, and the compressed copy fixes the site. The theorem below then identifies the lamps along the whole  $L$ -orbit of  $x_0$  in every corona representation.

**Theorem 11.14** (involutive collapse). *Let  $H$  be countable, let  $L \leq H$  be a subgroup with property (T), and let  $s \in H$  satisfy  $s\gamma s^{-1} \in L$  for every  $\gamma \in L$ . Then*

$$D_{\text{coll}}(L, s) \leq \text{Rad}_{\text{MF}}(H).$$

[Lean: GroupApproximation.ManuscriptExactWrappers.manuscriptInvolutiveCollapse]

Unwinding Definition 11.1: every corona representation  $\Theta$  of  $H$  satisfies  $\Theta([\gamma, k]) = 1$  for every witness  $k$  and every  $\gamma \in L$  — it identifies the entire  $L$ -conjugation orbit of each witness with the witness itself. No witness is marked in advance; the defect subgroup collects all of them at once.

*Proof.* Fix a witness  $k$  and suppose some corona representation separates a collapse commutator: coordinate unitary lifts (Lemma 2.2) give an operator-norm almost representation  $(U_{g,n})$  of  $H$  on blocks  $M_{d_n}(\mathbb{C})$  and a  $\gamma_0 \in L$  with  $\liminf_n \|U_{[\gamma_0, k], n} - 1\| > 0$ . We derive a contradiction.

*Step 1: exact commuting involution microstates.* For  $\gamma \in L$  the elements  $v_\gamma = \gamma k \gamma^{-1}$  have asymptotically commuting images that asymptotically square to 1, by (W1), (W3), and asymptotic multiplicativity. Correct them: for a unitary  $u$ , the spectral sign of its Hermitian part is an exact self-adjoint involution at operator distance at most  $\|u^2 - 1\|$  from  $u$ . Block compression along the finite commutative corner structure generated by previously corrected microstates preserves exact commutation. Inductively over an enumeration of  $L$  this produces exact, pairwise-commuting, self-adjoint involutions  $V_n(\gamma)$  with  $\|V_n(\gamma) - U_{v_\gamma, n}\| \rightarrow 0$ , covariant in operator norm:  $\|U_{g,n} V_n(\gamma) U_{g,n}^* - V_n(g\gamma)\| \rightarrow 0$  for each fixed  $g \in L$ , where  $g\gamma$  abbreviates the orbit index of  $gv_\gamma g^{-1}$ .

*Step 2: the rank displacement metric.* For commuting exact involutions  $u, w$  the displacement  $T = u - w$  is self-adjoint and satisfies  $T^3 = 4T$ , so  $T^2/4$  is an orthogonal projection and

$$\text{Tr } |u - w|^2 = 4 \text{rank}((u - w)^2/4) :$$

unnormalized Frobenius mass and rank are the same invariant on this locus. Set  $d_n(\gamma, \mu) = \text{rank}(V_n(\gamma) - V_n(\mu))$ , an integer-valued pseudometric. Orthogonal projections at operator distance less than 1 have equal rank, so the covariance of Step 1 makes the displacement eventually translation invariant:  $d_n(g\gamma, g\mu) = d_n(\gamma, \mu)$  for all large  $n$ , for each fixed triple  $(g, \gamma, \mu)$ .

*Step 3: generator normalization.* Property (T) makes  $L$  finitely generated; fix a finite symmetric generating set  $S$  forming a Kazhdan pair  $(S, \kappa)$ , and

put  $k_n = \sum_{a \in S} d_n(1, a)$ . Translation invariance and the triangle inequality for the rank pseudometric give  $d_n(1, \gamma) \leq |\gamma|_S k_n$  eventually, where  $|\cdot|_S$  is the word length. If  $k_n = 0$  along a subsequence, all generator involutions coincide there. By invariance the whole orbit then collapses to a single exact involution, so  $\|U_{[\gamma_0, k], n} - 1\| \rightarrow 0$  along that subsequence, against the separation. Hence  $k_n \geq 1$  for all large  $n$ .

*Step 4: the displacement almost-cocycle.* Flatten each block to the Euclidean space of matrices and set  $b_n(\gamma) = k_n^{-1/2}(V_n(\gamma) - V_n(1))$ . By Steps 2 and 3,  $\|b_n(\gamma)\|_F^2 \leq 4|\gamma|_S$  at every large stage, while the generators carry total mass exactly

$$(11) \quad \sum_{a \in S} \|b_n(a)\|_F^2 = 4$$

whenever  $k_n \geq 1$ . Conjugation  $x \mapsto U_{g,n} x U_{g,n}^*$  is a Frobenius isometry, and  $b_n$  satisfies the cocycle identity  $b_n(g\gamma) \approx b_n(g) + U_{g,n} b_n(\gamma) U_{g,n}^*$  up to a defect that is a sum of three covariance errors, each of rank  $O(k_n)$  and of vanishing operator norm. Since  $\text{Tr } |D|^2 \leq \text{rank}(D) \cdot \|D\|^2$ , the defect mass vanishes at the scale  $k_n$ . Every estimate below measures only differences of the involutions, never the involutions themselves.

*Step 5: bounded profile and an approximate primitive.* The limiting profile  $\gamma \mapsto \limsup_n \|b_n(\gamma)\|_F^2$  is finite on generators by (11). The Gaussian positive-definite functions  $e^{-t\|\cdot\|^2}$  of the almost-cocycle are almost invariant under the twisted action, and the spectral gap of the Kazhdan pair  $(S, \kappa)$  bounds the whole profile by its generator values — Delorme’s argument run directly at the level of limiting seminorms. With the profile bounded, a circumcenter of the displacement orbit, taken at the level of bounded sequences, produces one bounded sequence  $(w_n)$  that is an approximate primitive:  $b_n(a) \approx w_n - U_{a,n} w_n U_{a,n}^*$  in limiting normalized mass, simultaneously for any prescribed finite set of movers.

*Step 6: transport through the compressor.* By (W2) the compressed copy  $sLs^{-1}$  centralizes  $k$ , so the corrected microstates are eventually constant along it and the displacement vanishes on compressed elements. Including the compressed generators  $sas^{-1}$  among the prescribed movers of Step 5 therefore makes the primitive almost fixed by them. The pulled-back sequence  $u_n = U_{s,n}^* w_n U_{s,n}$  is then almost fixed by every generator  $a \in S$ , because microstates of  $as^{-1}$  and  $s^{-1}(sas^{-1})$  agree asymptotically. Now transport forward, exactly as in Theorem 3.1: compress  $u_n$  to the almost-fixed spectral subspace of the averaged adjoint action of  $S$  and apply the one-sided compression  $P \leq VPV^*$  at the rescaled mass. The sequence  $U_{s,n} u_n U_{s,n}^*$  returns  $w_n$  up to vanishing mass and is again almost fixed by every  $a \in S$ . Hence  $b_n(a) \approx w_n - U_{a,n} w_n U_{a,n}^*$  has small mass for every  $a \in S$  at a common large stage, contradicting (11).

Thus no corona representation separates any collapse commutator, and the kernel of a corona representation, being normal, contains all of  $D_{\text{coll}}(L, s)$ .  $\square$

*Remark 11.15* (the quantitative form). As with the transport theorem, Theorem 11.14 also has a fully quantitative single-stage proof: the limits above are replaced by explicit tolerances, the circumcenter by an approximate circumcenter of a near-optimal covering sequence, and the final contradiction is assembled from the displayed estimates at one sufficiently late coordinate. The commutation hypothesis (W3) is essential: it supplies the joint spectral calculus behind the rank identity of Step 2. Anticommuting Clifford coordinates have no joint spectral labels; their collapse mechanism is the central sign of Theorem A. At the corona level, Theorem 11.16 below removes the restriction to involutions: the spectral projections of a finite-order witness satisfy its hypotheses, and their collapse forces the collapse of the witness. The spectral-projection reduction is elementary functional calculus and is outside the formal development.

The witness of Theorem 11.14 is a group element. The same collapse holds for arbitrary projections of the corona.

**Theorem 11.16** (projection collapse). *Let  $H$  be countable, let  $L \leq H$  have property (T), let  $s \in H$  satisfy  $s\gamma s^{-1} \in L$  for every  $\gamma \in L$ , and let  $\Theta: H \rightarrow \mathcal{U}(\mathcal{Q})$  be a corona representation. Let  $p \in \mathcal{Q}$  be a projection such that  $p$  commutes with  $\Theta(s\gamma s^{-1})$  for every  $\gamma \in L$  and the projections  $\Theta(\gamma)p\Theta(\gamma)^*$ ,  $\gamma \in L$ , commute pairwise. Then  $p$  commutes with  $\Theta(L)$ .* [Lean: GroupApproximation.ProjectionCompressionCollapse.corona\_projection\_collapse]

*Proof.* The sign  $w = 1 - 2p$  is a self-adjoint unitary of  $\mathcal{Q}$ . The subgroup  $H' \leq \mathcal{U}(\mathcal{Q})$  generated by  $\Theta(H)$  and  $w$  is countable. Inside  $H'$ , the image  $\Theta(L)$  has property (T),  $\Theta(s)$  compresses it,  $w$  is an involution centralizing the compressed copy, and the  $\Theta(L)$ -conjugates of  $w$  commute because the conjugated projections do. So  $w$  is an involutive compression witness in  $H'$ , and Theorem 11.14 places every commutator  $[\Theta(\gamma), w]$ ,  $\gamma \in L$ , in  $\text{Rad}_{\text{MF}}(H')$ . The inclusion  $H' \rightarrow \mathcal{U}(\mathcal{Q})$  is itself a corona representation of  $H'$ , so it maps each of these commutators to the identity:  $[\Theta(\gamma), w] = 1$  in  $\mathcal{Q}$ , and  $w = 1 - 2p$  gives  $[\Theta(\gamma), p] = 0$ .  $\square$

The witness here need not come from a group element: the projection is an arbitrary element of the corona. The relative commutant  $\Theta(sLs^{-1})' \cap \mathcal{Q}$  therefore cannot exceed  $\Theta(L)' \cap \mathcal{Q}$  by a projection with commuting  $L$ -orbit.

**Corollary 11.17** (collapse computes the radical). *In the setting of Theorem 11.14, write  $q: H \rightarrow H/D_{\text{coll}}(L, s)$  for the quotient map. Then, unconditionally,*

$$\text{Rad}_{\text{MF}}(H) = q^{-1}(\text{Rad}_{\text{MF}}(H/D_{\text{coll}}(L, s))),$$

*and if  $H/D_{\text{coll}}(L, s)$  is MF, then  $\text{Rad}_{\text{MF}}(H) = D_{\text{coll}}(L, s)$ .* [Lean: GroupApproximation.ManuscriptExactWrappers.manuscriptCollapseRadicalReduction]

*Proof.* Combine Theorem 11.14 with Corollaries 11.5 and 11.4.  $\square$

In the second case  $H/D_{\text{coll}}(L, s)$  is the universal MF quotient of  $H$  (Proposition 11.3): the corona representation theory of  $H$  is computed exactly by the quotient.

**Theorem 11.18** (defect saturation). *In the setting of Theorem 11.14:*

- (1) *Every homomorphism  $f$  from  $H$  to a countable MF group maps every element of  $D_{\text{coll}}(L, s)$  to the identity.*
- (2) *If  $D_{\text{coll}}(L, s) = H$ , then  $\text{Rad}_{\text{MF}}(H) = H$ ; every homomorphism from  $H$  to a countable MF group is trivial; and if  $H$  is nontrivial, then  $H$  is not MF.*

[Lean: GroupApproximation.ManuscriptExactWrappers.manuscriptDefectSaturation]

*Proof.* (1) By Theorem 11.14 and Lemma 11.2,

$$f(D_{\text{coll}}(L, s)) \leq f(\text{Rad}_{\text{MF}}(H)) \leq \text{Rad}_{\text{MF}}(f(H)).$$

The subgroup  $f(H)$  of the MF target is MF (Lemma 11.9), so its radical is trivial by Proposition 11.3. (2) The radical statement is Theorem 11.14 with  $D_{\text{coll}}(L, s) = H$ ; triviality of every homomorphism follows from (1); and a nontrivial group with full radical is not MF by Proposition 11.3.  $\square$

A group as in (2) admits no nontrivial finite-dimensional unitary representation, no nontrivial MF quotient, and no nontrivial homomorphism to any countable residually finite or amenable group. The strategy — detect a defect analytically, then saturate it algebraically — develops a suggestion of Fournier-Facio (personal communication, 2026; cf. the Questions preamble of Section 12). Whether a small-cancellation construction produces a finitely presented group with saturated collapse or normal-Kazhdan defect is open.

**11.4 The reduced group algebra of  $E$**  Every MF algebra is stably finite [BK], so Theorem D exhibits a separable stably finite  $C^*$ -algebra that is not MF. Abstract examples follow from the failure of the Connes embedding problem [FGH]; the point here is concreteness: the algebra is the reduced group  $C^*$ -algebra of an explicit finitely presented group. Even for  $C_{\text{red}}^*(\text{SL}_3(\mathbb{Z}))$  and  $C_{\text{red}}^*(\text{SL}_4(\mathbb{Z}))$  the MF property is undecided; only the stronger PMF property is known to fail, for  $\text{SL}_4(\mathbb{Z})$  [MdlS].

**Theorem D** (the reduced group algebra of  $E$ ).  *$C_{\text{red}}^*(E)$  is a unital separable  $C^*$ -algebra which carries a faithful tracial state — and is therefore stably finite — but is not an MF algebra.*

[Lean: GroupApproximation.LiteralNonMFEndpoint.manuscriptTheoremD]

The canonical homomorphism  $\mathcal{B} \rightarrow E$  is injective. To see this, compose it with the homomorphism  $E \rightarrow W$  of Proposition 8.3; the composite is  $g \mapsto j(\bar{g})$ , where  $\bar{g}$  is the image of  $g$  under the surjection  $\mathcal{B} \rightarrow \bar{\Gamma}$  of Lemma 8.2. That surjection is an isomorphism by the classical identification of Remark 7.3: the twenty relators present the affine group  $\mathbb{Z}^3 \rtimes \text{SL}_3(\mathbb{Z})$ , and  $\bar{\Gamma}$  is that group in its matrix realization. The level-zero map  $j$  is injective, so the composite is injective. Since the infinite Kazhdan group  $\mathcal{B}$  is nonamenable, so is  $E$ . Hence  $C_{\text{red}}^*(E)$  is not nuclear [Lan]. The nuclear



form of the Blackadar–Kirchberg problem — whether every stably finite separable nuclear  $C^*$ -algebra is quasidiagonal — therefore remains open. This paragraph uses three classical literature inputs, used only here: the presentation identification, nonamenability of infinite Kazhdan groups, and nonnuclearity of the reduced algebra of a nonamenable group.

**Lemma 11.19.** *Let  $G_1$  be a countable discrete group.*

- (1) *The canonical tracial state  $\tau$  on  $C_{\text{red}}^*(G_1)$  is faithful.*
- (2) *Every unital  $C^*$ -algebra  $A$  with a faithful tracial state is stably finite: for every nonempty finite index set  $I$ , every isometry in  $M_I(A)$  is a unitary. In particular this holds in  $M_k(A)$  for every  $k \geq 1$ .* [Lean: GroupApproximation.ManuscriptExactWrappers.manuscriptFaithfulTraceAndStableFiniteness]

*Proof.* (1) Realize  $C_{\text{red}}^*(G_1) \subseteq B(\ell^2 G_1)$  as the norm closure of the span of the left translation unitaries  $\lambda_g$ . Let  $\rho_g$  be the right translation unitaries,  $(\rho_g \xi)(h) = \xi(hg)$ . Each  $\rho_g$  commutes with each  $\lambda_{g'}$ , hence with every element of  $C_{\text{red}}^*(G_1)$ . The canonical trace is  $\tau(x) = \langle x\delta_e, \delta_e \rangle$ . If  $\tau(x^*x) = 0$  then  $\|x\delta_e\| = 0$ , so  $x\delta_e = 0$ , and then  $x\delta_g = x\rho_{g^{-1}}\delta_e = \rho_{g^{-1}}x\delta_e = 0$  for every  $g \in G_1$ . The vectors  $\delta_g$  are total in  $\ell^2 G_1$ , so  $x = 0$ .

(2) On  $M_k(A)$  the functional  $\tau_k([x_{ij}]) = \frac{1}{k} \sum_i \tau(x_{ii})$  is a state with the trace property  $\tau_k(xy) = \frac{1}{k} \sum_{i,j} \tau(x_{ij}y_{ji}) = \tau_k(yx)$ , and it is faithful: the  $i$ -th diagonal entry of  $x^*x$  is  $\sum_j x_{ji}^*x_{ji}$ , so  $\tau_k(x^*x) = 0$  forces  $\tau(x_{ji}^*x_{ji}) = 0$ , hence  $x_{ji} = 0$ , for all  $i, j$ . If  $v \in M_k(A)$  satisfies  $v^*v = 1$ , then  $y := 1 - vv^*$  is a projection with  $\tau_k(y) = \tau_k(1) - \tau_k(vv^*) = \tau_k(1) - \tau_k(v^*v) = 0$ ; since  $y = y^*y$ , faithfulness gives  $y = 0$ , i.e.  $vv^* = 1$ .  $\square$

*Proof of Theorem D.*  $C_{\text{red}}^*(E)$  is unital and separable since  $E$  is countable; by Lemma 11.19 it carries a faithful tracial state and is stably finite. It is not an MF algebra by Theorem A.  $\square$

**11.5 A sofic non-MF group** The same quotient that proves  $w \neq 1$  also provides the sofic non-MF group:

**Theorem E** (a sofic non-MF group). *The Clifford witness  $W = \text{ClLamp}(X) \rtimes V$  of Proposition 8.3 is a finitely generated sofic group that is not MF; since sofic groups are hyperlinear,  $W$  is likewise a hyperlinear group that is not MF. In particular, neither soficity nor hyperlinearity implies the MF property.*

[Lean: GroupApproximation.NonMFImpact.witness\_sofic\_hyperlinear\_nonMF]

*Proof of Theorem E.* Write  $T = \varinjlim (\bar{\Gamma}, \bar{\alpha})$ , so  $V = T \rtimes \mathbb{Z}$ , and let  $B \leq V$  be the level-zero copy of  $\bar{\Gamma}$ . The image of  $\bar{\alpha}$  has finite index (at most eight, by the parity classes of the translation lattice). Thus every telescope level  $\bar{\Gamma}_n \leq T$  is commensurable with  $B$ . The stable letter conjugates  $B$  onto  $\bar{\alpha}(\bar{\Gamma})$ , so all of  $V$  commensurates  $B$ . Orbit–stabilizer now shows that every  $\bar{\Gamma}_n$ -orbit in  $X = V/B$  is finite.

Consider a finite subset of  $\text{ClLamp}(X) \rtimes T$ . Its  $T$ -coordinates lie in one level  $\bar{\Gamma}_n$ . Each of its lamp coordinates has finite  $\bar{\Gamma}_n$ -orbit: this holds for the central



sign and for each site lamp, hence for every Clifford word. The finitely many resulting orbits generate a finite  $\bar{\Gamma}_n$ -invariant subgroup  $L \leq \text{ClLamp}(X)$ , because  $\text{ClLamp}(X)$  is locally finite. Our original finite set therefore lies in  $L \rtimes \bar{\Gamma}_n$ , which is sofic (finite-by-sofic). Soficity is local over finite subsets, so  $\text{ClLamp}(X) \rtimes T$  is sofic.

Reassociating the semidirect products gives

$$W = \text{ClLamp}(X) \rtimes (T \rtimes \mathbb{Z}) \cong (\text{ClLamp}(X) \rtimes T) \rtimes \mathbb{Z}.$$

The final group is sofic because a split extension of a sofic group by  $\mathbb{Z}$  is sofic. Truncate the integer levels to  $\mathbb{Z}/N$  and let the shift act cyclically. Induce the kernel's sofic approximations over this finite level set. The cocycle identity that makes the finite-quotient case exact now fails only on the wrap-around boundary. For a window shifted by  $k$ , that boundary occupies at most  $2|k|/N$  of the levels, which vanishes as  $N$  grows. This is the  $\mathbb{Z}$ -extension case of the Elek–Szabó amenable-extension theorem [ES06]. Here  $\bar{\Gamma}$  is sofic: its six generators lie in  $\text{GL}_4(\mathbb{Z})$ , the resulting integral subgroup is residually finite through its congruence quotients, and it is isomorphic to  $\bar{\Gamma}$ .

The map  $E \rightarrow W$  of Proposition 8.3 is surjective: its image contains  $V$ , the root lamp  $c_o$ , and hence every lamp by transitivity of the  $V$ -action on  $X$ . Since  $E$  is generated by  $v_1, x, y, z, t, c$  (by the Tietze calculation above), their six images generate  $W$ . The map sends  $w \in \text{Rad}_{\text{MF}}(E)$  to the nontrivial sign  $\zeta$ ; functoriality (Lemma 11.2) makes  $\zeta \in \text{Rad}_{\text{MF}}(W)$ . Hence  $W$  is not MF.  $\square$

Failure of extension permanence. MF groups are not closed under extensions. More precisely,  $W$  fits into an exact sequence

$$1 \longrightarrow \text{ClLamp}(X) \longrightarrow W \longrightarrow V \longrightarrow 1$$

whose kernel and quotient are MF, while  $W$  is not. The concrete lamp kernel is locally finite, LEF, sofic, and MF, while the total witness is finitely generated and non-MF. [Lean: GroupApproximation.LiteralWitnessConsequences.literalWitness\_locallyFiniteKernel\_nonMF]

*Proof.* The Clifford lamp group is locally finite, hence MF by Lemma 11.11. The group  $V$  identifies with the subgroup of  $\text{GL}_4(\mathbb{Q})$  generated by  $\bar{\Gamma}$  and  $D$ : the telescope levels are  $D^{-n}\bar{\Gamma}D^n$ , and the determinant gives the  $\mathbb{Z}$ -coordinate. It is residually finite. At every odd modulus the determinant 8 of the dilation is invertible, so the congruence reduction of the integral affine model descends to conjugation by the dilation and extends over the telescope and its shift. These reductions separate any nontrivial element; compare Mal'cev's theorem [Mal] for finitely generated linear groups. Thus  $V$  is MF by Lemma 11.10, whereas  $W$  is not MF by the preceding construction.  $\square$

An exact stably finite non-MF algebra. The same witness gives a concrete separable exact example:  $C_{\text{red}}^*(W)$  is exact, has a faithful tracial state, is stably finite, and is not MF. Indeed, the locally finite lamp kernel is amenable and hence exact, while the linear group  $V \leq \text{GL}_4(\mathbb{Q})$  is exact [GHW]. Exact

groups are closed under extensions [KW2], so  $W$  is exact; for discrete groups this is equivalent to exactness of the reduced group algebra [KW1]. The canonical trace and stable finiteness follow as in Lemma 11.19. Finally, an MF embedding of  $C_{\text{red}}^*(W)$  would restrict to a faithful norm-corona model of the canonical group unitaries, contradicting that  $W$  is not MF. All parts of this paragraph except exactness are also verified in Lean; the present formal library has no definition of exact  $C^*$ -algebras.

A simple sofic envelope. There is a countable simple sofic group  $S_{\text{simp}}$  with

$$\text{Rad}_{\text{MF}}(S_{\text{simp}}) = S_{\text{simp}}.$$

Equivalently, every homomorphism from  $S_{\text{simp}}$  to an MF group is trivial.

*Proof.* By Elek–Szabó, every countable sofic group embeds in a countable simple sofic group [ES05, Theorem 1]. Embed  $W$  in such a group. Functoriality of the MF radical sends  $\zeta$  to a nontrivial element of the MF radical of the simple group. That radical is normal, so it is the whole group. Every homomorphism to an MF group kills the MF radical.  $\square$

**11.6 Marked-limit closure** The explicit clopen cylinder constructed in the next subsection is an instance of a general phenomenon: at every fixed number of marked generators, the MF property is closed under limits of marked groups, and its failure always has a finite certificate.

**Theorem 11.20** (the MF locus is closed). *Fix  $k \geq 1$  and consider the space of  $k$ -marked groups — quotients of the free group  $F_k$ , topologized so that a basic clopen set fixes the truth of finitely many equations among words in the generators.*

- (1) *The marked groups with MF quotient form a closed set.* [Lean: GroupApproximation.MarkedGroupSpace.isClosed\_operatorMFLocus]
- (2) *Equivalently, the non-MF marked groups form an open set.* [Lean: GroupApproximation.MarkedGroupSpace.isOpen\_compl\_operatorMFLocus]
- (3) *Every non-MF  $k$ -marked group admits a finite-radius certificate: there is a radius  $R$  such that every  $k$ -marked group whose relations agree with it on the reduced words of length at most  $R$  is non-MF.* [Lean: GroupApproximation.MarkedGroupSpace.exists\_wordBall\_cylinder\_subset\_compl\_operatorMFLocus]

*Proof.* A failure of MF for the quotient is a failure of normalized local approximation, and such a failure is witnessed on one finite subset of the quotient. Expressing the finitely many multiplications and inequalities of that subset by words in the marked generators produces finitely many relation constraints; every marked group agreeing on those constraints transports the failed local-model problem verbatim, so an entire basic clopen neighborhood fails MF. Enclosing the constraint words in a reduced-word ball gives the radius form.  $\square$

Thus failure of MF is always certified by one finite ball of the marked Cayley diagram. The next subsection computes such a certificate explicitly for  $E$ .

**11.7 A finite universal Horn obstruction** Theorem A can be read as one first-order fact about all MF groups at once: a finite quasi-identity, that is, a single universal implication between finitely many equations (a universal Horn sentence). Write  $R$  for the forty-one relators of (10), regarded as words in eight variables, and  $w$  for the marked word in those variables.

**Proposition 11.21** (finite quasi-identity). *Every MF group  $G$  satisfies the finite universal Horn sentence*

$$\forall x_1, \dots, x_8: \bigwedge_{r \in R} r(x_1, \dots, x_8) = 1 \implies w(x_1, \dots, x_8) = 1,$$

while the canonical generating tuple of  $E$  satisfies every relator in  $R$  and has  $w \neq 1$ . [Lean: GroupApproximation.LiteralUniversalHorn.manuscriptLiteralUniversalHorn]

*Proof.* A tuple satisfying the relators induces a homomorphism  $E \rightarrow G$  sending the marked word of  $E$  to  $w(x_1, \dots, x_8)$ . If  $G$  is MF, that image is trivial:  $w \in \text{Rad}_{\text{MF}}(E)$  by Theorem A, and the radical is functorial (Lemma 11.2). The canonical tuple of  $E$  satisfies  $R$  by definition, and its marked word is nontrivial by Theorem A(1).  $\square$

**Corollary 11.22** (a basic clopen set of non-MF marked groups). *In the space of eight-generator marked groups — quotients of the free group  $F_8$ , topologized so that a basic clopen set fixes the truth of finitely many equations among words in the generators — the marked groups satisfying all relators of  $R$  and  $w \neq 1$  form a nonempty clopen set consisting entirely of non-MF groups.* [Lean: GroupApproximation.LiteralMarkedCylinder.literal\_nonempty\_clopen\_nonMF\_cylinder]

*Proof.* Both conditions constrain finitely many values of the marked multiplication, so the set is clopen; it contains the marked group  $E$  by Theorem A(1), and Proposition 11.21 excludes every MF point.  $\square$

Thus the obstruction of Theorem A is finite, local, and locally constant on a basic clopen neighborhood in the marked-group topology.

**11.8 Undecidability of MF recognition** Lemma 11.9 and Theorem A make MF a Markov property of finitely presented groups: the trivial group is MF, and by subgroup heredity the finitely presented group  $E$  embeds in no MF group. The Adian–Rabin theorem [Ra] therefore applies.

**Corollary 11.23** (undecidability). *There is no algorithm that decides, from a finite presentation, whether the presented group is MF. In the effective form of the Adian–Rabin construction, the set of finite presentations of non-MF groups is not even recursively enumerable.*

This corollary carries one classical literature input, used only here: the Adian–Rabin construction itself [Ra], which we have not formalized. Everything group-theoretic that feeds it is machine-checked — subgroup heredity of MF and the finitely presented non-MF group — [Lean: GroupApproximation.MarkovMFConsequences.operatorMF\_subgroup\_hereditary] [Lean: GroupApproximation.MarkovMFConsequences.exists\_finitelyPresented\_nonOperatorMF] and so is the computability-theoretic pullback: from any computable Adian–Rabin reduction, undecidability and the failure of recursive enumerability on the negative side follow formally. [Lean: GroupApproximation.MarkovMFConsequences.operatorMF\_recognition\_undecidable] Theorem 11.20 certifies single non-MF groups by finite balls; the corollary says no algorithm classifies the positive and negative cases from presentations.

The phenomenon also has infinite multiplicity, in two senses that need no new analysis. The groups  $E \times \mathbb{Z}^k$  for  $k \geq 0$  are finitely presented, pairwise nonisomorphic — their abelianizations have distinct torsion-free ranks — and non-MF, each containing  $E$  (Lemma 11.9). Beyond finite presentation, pairing  $E$  with the continuum of pairwise nonisomorphic two-generator groups of B. H. Neumann [Ne] yields  $2^{\aleph_0}$  pairwise nonisomorphic finitely generated non-MF groups: each product  $E \times N$  is non-MF, and a fixed countable group  $X$  satisfies  $X \cong E \times N$  for at most countably many of the continuum many isomorphism types  $N$  — each such  $N$  embeds in  $X$  as a finitely generated subgroup, a countable group has only countably many finitely generated subgroups, and the types  $N$  are pairwise nonisomorphic. A map with countable fibres from a continuum-sized family has continuum-sized image. These two paragraphs are elementary group theory on top of the machine-checked heredity lemma and are not separately formalized.

## 12 LIMITATIONS OF THE OPERATOR-NORM METHOD

The proof is specific to the operator norm. If multiplication is controlled only in normalized Hilbert–Schmidt norm, the induced adjoint operators on the Hilbert–Schmidt spaces need not be close in operator norm. Both transport proofs require exactly that operator-norm control: the ultraproduct proof needs it to obtain an honest representation on the ultraproduct, and the finite-stage proof of Appendix B needs it for the spectral estimates. Normalized Hilbert–Schmidt control therefore does not supply the transport used here.

This is a limitation of the method, not a tracial nonapproximation theorem for  $E$ . Alekseev–Thom ask whether commutants arising in sofic approximations of Kazhdan groups can be recovered from finite-dimensional coordinate subalgebras [AT, Open Problem 6.2]. Such a coordinate description is separate from the operator-norm theorem proved here.

Finite dimensionality of the coordinates is load-bearing in a second, sharper sense: it cannot be traded for stable finiteness of the coordinate algebras together with faithful traces. In a general faithfully traced unital  $C^*$ -algebra

the transport argument would act on the trace Hilbert space by the left and right regular actions, and the von Neumann algebra these two actions generate can fail every finiteness property — for an infinite-dimensional  $\text{II}_1$  factor in standard form it is all bounded operators on the trace space. Constant coordinates  $A_n = C_{\text{red}}^*(E)$  refute any such generalization outright: the left regular representation embeds  $E$  exactly and injectively into the corresponding corona, the element  $c$  centralizes the Kazhdan base there, and conjugation by  $t$  produces  $tct^{-1}$ , whose commutator with the base is precisely the nontrivial defect that Theorem A detects. A faithfully-traced coordinate transport theorem would collapse that commutator; no such theorem can hold.

**Questions** The finite-normal criterion has a precise torsion limitation: every finite subgroup of a torsion-free group is trivial. [Lean: GroupApproximation.finiteSubgroup\_eq\_bot\_of\_isMulTorsionFree] By Theorem 4.6, a normal property-(T) subgroup of the defect subgroup obstructs corona representations without any torsion. What remains for a torsion-free example is group-theoretic: a torsion-free finitely presented group whose defect subgroup contains a normal subgroup with property (T). Fournier-Facio, who removed the analogous torsion obstacle for nonsocicity [FFF], has suggested (personal communication, 2026; cf. Remark 5.2) that a small-cancellation construction should produce such a group.

- (1) Is  $E/\langle w \rangle$  MF? By Corollary 11.4, a positive answer would compute  $\text{Rad}_{\text{MF}}(E) = \{1, w\}$  exactly. There is a concrete intermediate test. Theorem B sends  $u = [tct^{-1}, v_1]$  to the identity in every finite-dimensional representation, whereas the image of  $u$  in  $W/\langle \zeta \rangle$  is nontrivial: it is the product of the two distinct lamp basis elements at  $\tau o$  and  $j(\bar{v}_1)\tau o$ . Thus proving  $W/\langle \zeta \rangle$  MF would already show  $u \notin \text{Rad}_{\text{MF}}(E)$ ; by Theorem B, no such detecting model can factor through finite quotients or exact finite-dimensional representations.
- (2) Is there a torsion-free finitely presented non-MF group? The analytic obstruction no longer requires torsion (Theorem 4.6); the remaining input is group-theoretic. A construction along the lines suggested above — strengthening Hull’s common-quotient technique so that the target elements are chosen inside a given infinite normal subgroup of a torsion-free acylindrically hyperbolic group, applied to Fournier-Facio’s skeleton — has been carried out and internally audited at paper level, and yields the stronger conclusion of a torsion-free finitely presented group equal to its own MF radical, so that every homomorphism to every norm matrix corona is trivial. That argument rests on the Hull–Osin small-cancellation theory and is neither formalized nor part of the present development; until it is verified independently we record the question as resolved only modulo that construction. By Theorem 11.18, a group whose collapse or normal-Kazhdan defect saturates admits no nontrivial homomorphism to any countable MF group at all.

- (3) Is  $E$  sofic, or hyperlinear? The adjoint spectral estimates require operator-norm multiplicative control, and neither property is decided by the results above.
- (4) Does MF imply hyperlinearity? A faithful embedding into a norm matrix corona need not make any coordinate trace faithful, so the defining MF representation does not immediately supply a hyperlinear model. This question concerns the CDE norm-corona convention fixed above; for stronger trace-controlled MF conventions the implication is built into the definition.
- (5) By Proposition 11.7, every family of theorems placing subgroups in the MF radical generates, by iteration through quotients, a closure contained in  $\text{Rad}_{\text{MF}}$ . For which groups do the rules of Theorems 4.2, 4.6, 11.14, and 11.16 compute the radical exactly?

#### A PROOF OF THE MF EQUIVALENCES

An *operator-norm local model* for a countable group  $G$ , a finite set  $F \subseteq G$ , a separation constant  $\delta > 0$ , and a tolerance  $\varepsilon > 0$  is a map  $V : G \rightarrow \mathcal{U}(r)$ , for some positive dimension  $r$ , such that

$$\begin{aligned} \|V(g)V(h) - V(gh)\| &\leq \varepsilon \quad (g, h \in F), \\ \|V(g) - V(h)\| &\geq \delta \quad (g \neq h \in F). \end{aligned}$$

The local formulation in Proposition 1.1 asks for such a model with  $\delta = 1$  for every finite  $F$  and every  $\varepsilon > 0$ .

*Proof of Proposition 1.1.* Lemma 2.2 identifies the unitary-sequence quotient with the unitary group of the  $C^*$ -quotient. Replacing an arbitrary finite coordinate set by a standard set of the same cardinality merely conjugates that coordinate by a permutation unitary. This proves the first equivalence once the dimension convention is normalized.

Suppose first that  $\rho : G \rightarrow \text{U}_{\text{cor}}((d_n))$  is injective, and choose unitary lifts  $u_n(g)$  of  $\rho(g)$ . Fix a finite set  $F \subseteq G$  and  $\varepsilon > 0$ . For each ordered pair  $g \neq h$  in  $F$ , injectivity says that  $(u_n(g)^{-1}u_n(h))_n$  does not converge to the identity in operator norm. Hence there are  $\delta_{g,h} > 0$  and arbitrarily large coordinates at which

$$\|u_n(g) - u_n(h)\| \geq \delta_{g,h}.$$

Choose an integer  $N_{g,h}$  with  $8 < N_{g,h} \delta_{g,h}^2$ . An explicit tensor-power estimate then applies: whenever two unitaries are  $\delta_{g,h}$ -separated, some tensor power of exponent  $p \leq N_{g,h}$  separates them by more than 1 — amplifying a spectral gap of the relative unitary until it crosses the unit circle takes at most  $N_{g,h}$  steps once  $8 < N_{g,h} \delta_{g,h}^2$ . The quotient homomorphism law makes all multiplicative defects for pairs in  $F$  arbitrarily small at every sufficiently large coordinate. Choose, for each  $g \neq h$ , a coordinate with the displayed separation and with every required defect at most  $\varepsilon/N_{g,h}$ . Then the  $p$ th tensor power separates the designated pair by more than 1, while the tensor-product telescoping identity bounds each new defect by  $p\varepsilon/N_{g,h} \leq \varepsilon$ . Finally

take the block sum of these finitely many pair-specific models. Operator norms of block diagonal matrices are maxima, so every pair in  $F$  is separated by at least 1 and every required product has defect at most  $\varepsilon$ . This gives the normalized local model.

Conversely, local models with separation constant 1 for all finite sets and tolerances make the group *weakly* MF. Choose models along an exhaustion with errors tending to zero. Then each fixed pair  $g, h$  is eventually represented multiplicatively, and each fixed pair  $g \neq h$  is eventually separated by 1. Hence the classes define an injective homomorphism into the unitary-sequence quotient, with no condition on the dimension sequence. Weak MF in turn makes the norm-MF radical of Definition 11.1 trivial, and the block-sum diagonalization in the proof of Proposition 11.3(2) then assembles the detecting sequences into a single faithful corona representation. This weak-to-corona implication is thus obtained from the universal-quotient construction, not from a separate exhaustion.

Finally, we normalize dimensions. Given positive dimensions  $d_n$ , replace the  $n$ th coordinate model by the cumulative block sum on  $X_0 \oplus \cdots \oplus X_n$ , where  $X_k$  is the  $k$ th original model space. Embed the  $n$ th original model in the newest block and put identity blocks in all earlier positions. The dimensions  $D_n = d_0 + \cdots + d_n$  increase strictly. Products, multiplicative defects, and pairwise operator-norm distances are unchanged, because block-diagonal operator norms are maxima and every earlier block is the same identity for all group elements; in particular no approximation error from earlier coordinates is carried forward. Thus the construction preserves and reflects the quotient relation and injectivity. Forgetting strict increase proves the converse.  $\square$

## B THE FINITE-STAGE TRANSPORT PROOF

This appendix proves Theorem 3.1 a second time, quantitatively at each single coordinate  $n$  and with no ultrafilter. Corollary 4.4 and Section 12 refer to this proof.

*Finite-stage proof of Theorem 3.1.* Property (T) provides a finite symmetric generating set  $S \subseteq \Gamma$  with  $1 \in S$  and a Kazhdan constant  $\kappa \in (0, 1]$  for the pair  $(S, \kappa)$ . The adjoint model is that of the printed proof: each group element acts by conjugation,  $\text{Ad } U_n(g) \xi = U_n(g) \xi U_n(g)^*$ , on  $L^2(M_{d_n}(\mathbb{C}), \text{tr}_{d_n})$ , and operator-norm almost multiplicativity of  $U_n$  makes  $g \mapsto \text{Ad } U_n(g)$  an operator-norm asymptotic representation on these Hilbert spaces.

*The almost-fixed spectral subspace.* Let  $H_n$  be the Hermitian part of the average  $\frac{1}{|S|} \sum_{s' \in S} \text{Ad } U_n(\iota(s'))$ , fix the threshold

$$\theta = \frac{1}{2} \left( \left( 1 - \frac{\kappa^2}{4|S|} \right) + 1 \right) < 1,$$

and let  $P_n$  be the spectral projection of  $H_n$  for the part of the spectrum above  $\theta$ . Two spectral estimates tie  $P_n$  to the group.



- (i) *Displacement.* For every  $\gamma \in \Gamma$ ,  $\|(\text{Ad } U_n(\iota(\gamma)) - 1)P_n\| \rightarrow 0$ . Indeed, a vector whose average  $S$ -energy exceeds  $\theta$  is almost fixed by each generator, because  $(S, \kappa)$  is a Kazhdan pair and  $\theta$  exceeds  $1 - \kappa^2/(4|S|)$ . It is then almost fixed by every word in  $S$ . The projection  $P_n$  is the finite-stage substitute for the Kazhdan projection.
- (ii) *Spectral concentration.* Write  $\xi_n \in L^2(M_{d_n}(\mathbb{C}), \text{tr}_{d_n})$  for the vector  $x_n$ . The commutator hypothesis says each  $\text{Ad } U_n(\iota(s'))$  almost fixes  $\xi_n$ , so the residual  $\xi_n - H_n \xi_n$  is Hilbert–Schmidt null. A Cauchy–Schwarz energy estimate then bounds the mass of  $(1 - P_n)\xi_n$  by the residual and the uniform bound on  $\|x_n\|$ . Hence  $\xi_n$  is asymptotically concentrated in  $\text{ran } P_n$ .

*Reversing the one-sided projection estimate.* Put  $T_n = \text{Ad } U_n(s)$  and  $Q_n = T_n P_n T_n^*$ , a projection of the same rank as  $P_n$ . The conjugation relation enters exactly once: for  $\gamma \in \Gamma$  the element  $s^{-1}$  pulls the almost-fixedness of a vector of  $\text{ran } P_n$  under  $\iota(\Gamma) \supseteq s\iota(\Gamma)s^{-1}$  back to almost-fixedness under each  $\iota(s')$ ,  $s' \in S$ . By the concentration estimate, the pulled-back vector is again almost in the subspace. Thus  $\text{ran } P_n$  is almost contained in its conjugate,  $\|(1 - Q_n)P_n\| \rightarrow 0$  — the finite-stage form of the one-sided inclusion  $P \leq VPV^*$ . Since  $\text{rank } P_n = \text{rank } Q_n$ , the equal-rank reversal estimate — for projections  $p, q$  of equal finite rank,  $\|(1 - q)p\| \leq \varepsilon < 1$  implies  $\|(1 - p)q\| \leq \varepsilon/\sqrt{1 - \varepsilon^2}$  — reverses the containment:

$$\boxed{\|(1 - P_n)Q_n\| \rightarrow 0.}$$

This is the middle implication of the display at the head of Section 3, and it is the finite-dimensional substitute for finiteness of a norm ultraproduct: a one-sided compression cannot enlarge the subspace.

*Combining the estimates.* Fix  $\gamma$  and  $\varepsilon > 0$ , let  $M \geq 1$  bound  $\|x_n\|$ , and put  $q = \min(1, \varepsilon/(40(M^2 + 1)))$ . For all large  $n$  the three estimates hold at tolerance  $q$ :

$$\|(1 - P_n)\xi_n\|_2^2 \leq q, \quad \|(1 - P_n)Q_n\| \leq q, \quad \|(A_n - 1)P_n\| \leq q,$$

where  $A_n = \text{Ad } U_n(\iota(\gamma))$ . Decompose the transported vector along  $P_n$  and  $Q_n$ :

$$T_n \xi_n = P_n Q_n T_n \xi_n + (1 - P_n)Q_n T_n \xi_n + (1 - Q_n)T_n \xi_n.$$

Since  $Q_n = T_n P_n T_n^*$  and  $T_n$  is conjugation by a unitary,

$$\|(1 - Q_n)T_n \xi_n\|_2 = \|T_n(1 - P_n)\xi_n\|_2 = \|(1 - P_n)\xi_n\|_2 \leq q^{1/2};$$

the middle term obeys  $\|(1 - P_n)Q_n T_n \xi_n\|_2 \leq \|(1 - P_n)Q_n\| \|\xi_n\|_2 \leq qM$ , and the displacement estimate moves the first term by at most  $q \|\xi_n\|_2 \leq qM$ . Applying  $A_n - 1$ , whose operator norm is at most 2, to the last two terms and using the displacement bound on the first gives

$$\|(A_n - 1)T_n \xi_n\|_2^2 \leq (qM + 2qM + 2q^{1/2})^2 \leq 18q^2 M^2 + 16q \leq \varepsilon.$$

Since  $T_n \xi_n$  is the vector of  $U_n(s)x_n U_n(s)^*$ , this proves the asserted convergence — with an explicit rate in terms of  $\kappa$ ,  $|S|$ ,  $M$ , and the multiplicative defects of  $U_n$  on the finitely many group elements involved.  $\square$

## C THE MAXIMAL GROUP $C^*$ -ALGEBRA

The familiar construction of the maximal group  $C^*$ -algebra completes the group ring over all unitary representations at once, and “all” is a class rather than a set. This appendix carries out the standard size-honest version, relative to a fixed set-theoretic universe  $\mathfrak{U}$ ; the result is the usual maximal group  $C^*$ -algebra. For a  $\mathfrak{U}$ -small group  $H$ , define  $C_{\max, \mathfrak{U}}^*(H)$  to be the closed  $*$ -subalgebra generated by the diagonal copy of  $H$  in the bounded product of all its nontrivial unitary representations on  $\mathfrak{U}$ -small unital  $C^*$ -algebras. We suppress the subscript  $\mathfrak{U}$  and write  $u_h$  for the canonical unitary associated with  $h \in H$ .

**Proposition C.1** (universe-relative maximal group algebra). *The canonical map  $h \mapsto u_h$  is injective. For every  $\mathfrak{U}$ -small unital  $C^*$ -algebra  $B$  and every homomorphism  $\rho: H \rightarrow \mathcal{U}(B)$ , there is a unique unital  $*$ -homomorphism  $C_{\max, \mathfrak{U}}^*(H) \rightarrow B$  mapping  $u_h$  to  $\rho(h)$  for every  $h \in H$ .* [Lean: GroupApproximation.manuscriptUniverseRelativeMaximalGroupCStar]

For a nontrivial target, projection to its coordinate gives the extension; for a trivial target, the map is unique. Density of the generated  $*$ -subalgebra gives uniqueness, while the left-regular coordinate proves injectivity.

The maximal and reduced algebras separate already at the level of finiteness. The algebraic mechanism is machine-checked in full:

**Proposition C.2** (proper compression forbids finiteness). *Let  $A$  be a unital  $C^*$ -algebra (a unital star ring suffices) containing a projection  $p$  and a unitary  $u$  such that the conjugate  $upu^*$  absorbs  $p$  on both sides and  $upu^* \neq p$ . Then  $A$  contains an isometry that is not a unitary; consequently  $A$  is not stably finite and carries no faithful tracial state.* [Lean: GroupApproximation.ProperProjectionCompression.isometry\_not\_isUnit] [Lean: GroupApproximation.ProperProjectionCompression.not\_isStablyFiniteRing] [Lean: GroupApproximation.ProperProjectionCompression.no\_faithfulTracialState]

*Remark C.3* (the maximal algebra under strict Kazhdan compression). For a group carrying a strict Kazhdan compression  $t\Gamma t^{-1} \subsetneq \Gamma$  — the base of  $E$  is the running example — the Kazhdan projection  $p_\Gamma$  constructed in Section 3 exists in the maximal algebra, conjugation by  $u_t$  dominates it, and the domination is strict: at the quasi-regular coordinate  $\ell^2(G/t\Gamma t^{-1})$  the point mass at the base coset is fixed by the compressed subgroup and moved by every element outside it, so the two projections differ there. [Lean: GroupApproximation.QuasiRegularWitness.baseVector\_apply\_base] By Proposition C.2 the maximal algebra of such a group is then not stably finite, carries no faithful trace, and in particular is neither residually finite-dimensional nor MF. The assembly of the Kazhdan projection with the coordinate spectral gap is stated here at paper level and is not part of the formal development.

The contrast with Theorem D is the point of this remark: for the reduced algebra of  $E$  the canonical trace is faithful and the algebra is stably finite, so its failure of MF is not a failure of finiteness; for the maximal algebra, infiniteness is forced by the strict compression before any MF considerations arise.

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